

Neuronal Detection Triggers Systemic Digestive Shutdown in Response to Adverse Food Sources

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This **important** study investigates how signals from the nervous system can influence the response to different food sources. To demonstrate the role of specific neuronal and intestinal regulators in sensing food quality and modulating digestion, the authors present evidence through a combination of genetic screening, RNA-seq analysis, and functional studies. These findings shed light on an adaptive strategy to integrate food perception with physiological responses, with a mix of **solid** and **convincing** evidence supporting the work.

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Abstract

The ability to sense and adapt to adverse food conditions is essential for survival across species, but the detailed mechanisms of neuron-digestive crosstalk in food sensing and adaptation remain poorly understood. This study identifies a novel mechanism by which animals detect unfavorable food sources through neurons and initiate a systemic response to shut down digestion, thus safeguarding against potential harm. Specifically, we demonstrate that NSY-1, expressed in AWC neurons, detects *Staphylococcus saprophyticus* (SS) as an unfavorable food source, prompting the animal to avoid and halt digestion of SS. Upon detection, the animals activate the AWC^{OFF} neural circuit, leading to a systemic digestive shutdown, which is mediated by NSY-1-dependent STR-130. Additionally, NSY-1 mutation triggers the production of insulin peptides, including INS-23, which interact with the DAF-2 receptor to modulate SS digestion and affects the expression of intestinal BCF-1. These findings uncover a crucial survival strategy through neuron-digestive crosstalk, where the NSY-1 pathway in AWC neurons orchestrates food evaluation and initiates digestive shutdown to adapt effectively to harmful food sources.

Introduction

Food is a source of essential nutrients and also poses a risk of lethal toxins and pathogens. Animals, including humans, must respond to various food sources to ensure survival. The ability to detect and adapt to adverse food conditions is crucial for the survival of many species. Various sensory mechanisms have evolved to monitor food quality by detecting beneficial and harmful substances, include olfactory (Fiala, 2007 [↗](#); McLachlan et al., 2022 [↗](#); Sengupta et al., 1996 [↗](#)), gustatory (Avery et al., 2021 [↗](#); Hukema et al., 2006 [↗](#); Scott, 2018 [↗](#)), and gut chemosensory systems (Bargmann, 2006 [↗](#)). Food allergies serve as a biological food quality control system, offering protection and benefits by promoting food avoidance behavior (Florsheim et al., 2021 [↗](#)). Research by Plum et al. (Plum et al., 2023 [↗](#)) and Florsheim et al. (Florsheim et al., 2023 [↗](#)) provides evidence that the immune system's allergic response communicates with the brain in mice, leading to food avoidance. This avoidance behavior acts as a defense strategy, reducing the risk of exposure to harmful substances, including allergens.

The digestive system functions by transporting food through the gastrointestinal (GI) tract, where it is broken down into molecules that can be absorbed and utilized by the body's cells. Thus, shutdown of digestion may serve as a mechanism for eliminating indigestible or harmful substances, acting as a protective system in animals to avoid adverse food. Despite this, the interaction between neuronal food detection and intestinal digestion, particularly in assessing and adapting to harmful food, remains inadequately understood.

The free-living nematode *Caenorhabditis elegans* thrives in organic-rich environments where it encounters a variety of microorganisms as food (Felix and Braendle, 2010 [↗](#); Samuel et al., 2016 [↗](#); Schulenburg and Felix, 2017 [↗](#)). *C. elegans* has evolved mechanisms to sense bacterial presence and food quality, which influence its feeding behaviors and digestive processes to adapt to its environment. Previous research has identified heat-killed *E. coli* as low-quality food, which the nematode avoids using its food-quality evaluation systems, such as the FAD-ATP-TORC1-ELT-2 pathway (Qi et al., 2017 [↗](#)) and the UPR^{ER}–immunity pathway (Liu et al., 2024 [↗](#)). Additionally, we have found that certain bacteria, like *Staphylococcus saprophyticus* (SS), are classified as inedible, leading to shutting down digestion and stunted growth in the nematodes (Geng et al., 2022 [↗](#)). However, we still need to determine whether SS represents harmful food for *C. elegans* and how the nematode senses SS and shuts down digestion to reject it. It is hypothesized that *C. elegans* may detect or assess inedible food, such as SS, and subsequently halt digestion as a survival strategy. This suggests that the cooperation between food sensing and digestive systems could form a systemic food quality control mechanism in animals, aimed at minimizing the adverse effects of harmful food.

In this study, we developed a model in *C. elegans* to investigate responses to harmful food and explored how the nematodes sense and avoid ingesting such food by shutting down their digestive systems. We identified a food quality control mechanism involving communication between neurons and the digestive system. This mechanism functions as a defense strategy to minimize the adverse effects of harmful food environments on the animals.

Results

Shutting Down Digestion as a Protective Mechanism for Survival in Larval *C. elegans*

Previous studies have shown that *Caenorhabditis elegans* cannot digest *Staphylococcus saprophyticus* (SS), which prevents their development (**Figure 1A**). In natural environments, *C. elegans* rely on various bacteria for nutrition and growth.

However, SS is not a viable food source for *C. elegans*. Larval arrest, such as the dauer stage, serves as an adaptive mechanism for survival under unfavorable conditions, including limited food availability or extreme temperatures^{8,9}. We hypothesize that *C. elegans* can sense or evaluate inedible food, such as SS, and subsequently shut down their digestion to arrest development as a protective survival strategy (**Figure 1B**).

Our previous research demonstrated that *C. elegans* can assess and avoid low-quality food, like heat-killed *E. coli*, to adapt to nutrient-deficient conditions (Liu et al., 2024; Qi et al., 2017). To determine if *C. elegans* also detect SS as an unfavorable food source, we conducted two behavioral assays: food dwelling/avoidance and food choice (Qi et al., 2017). In the food dwelling/avoidance assay, larval-stage animals exhibited strong discrimination against SS compared to the standard food, OP50 (**Figure 1C**). In the food choice assay, the animals preferred OP50 over low-quality food such as heat-killed *E. coli* (Figure 1—figure supplement 1A) or SS (**Figure 1D**). However, they could not distinguish between heat-killed *E. coli* and SS (Figure 1—figure supplement 1B). These results suggest that SS acts as an unfavorable food that *C. elegans* can detect and avoid.

In response to starvation, L1 larvae can enter a state of developmental arrest, pausing their growth to survive (Baugh and Sternberg, 2006). To test whether *C. elegans* shut down digestion of SS as a protective strategy upon sensing unfavorable food, we performed a survival assay on L1 larvae fed with SS. We found that larvae unable to digest SS still survived under SS feeding conditions (Figure 1—figure supplement 1C), similar to larvae under L1 starvation. This suggests that shutting down digestion may be a protective mechanism in larvae under SS feeding conditions.

Previously, we observed that activating larval digestion with heat-killed *E. coli* or *E. coli* cell wall peptidoglycan (PGN) enabled the digestion of SS as food (Hao et al., 2024). Additionally, when animals reached the L2 stage by feeding normal OP50 diet, they could utilize SS as a food source to support growth (Figure 1—figure supplement 1D). These findings suggest that once digestion is activated (via *E. coli* components or L2-stage maturation), worms gain the capacity to process SS as a viable food source, abolishing SS-induced growth impairment (Hao et al., 2024) (Figure 1—figure supplement 1D).

If SS is indeed an unfavorable or toxic food for *C. elegans*, digesting it could result in physiological defects. We measured the lifespan of L4 stage animals fed with SS or OP50. We found that SS consumption shortened their lifespan (**Figure 1E-F**), indicating a cost associated with digesting unfavorable or toxic food.

In conclusion, our data suggest that larval-stage *C. elegans* can sense and evaluate SS as an unfavorable food source, leading to the shutdown of digestion to avoid consumption, thereby protecting them and allowing adaptation to an unfavorable food environment.

C. elegans Sense SS and Shut Down Digestion through NSY-1

We speculated that key factors in *C. elegans* are involved in sensing *Staphylococcus saprophyticus* (SS) and shutting down its digestion. If these factors are mutated, the animals would fail to detect SS as an unfavorable food source and would utilize it (Figure 2—figure supplement 1A). To identify these factors, we conducted an unbiased forward genetic screen to find mutant animals

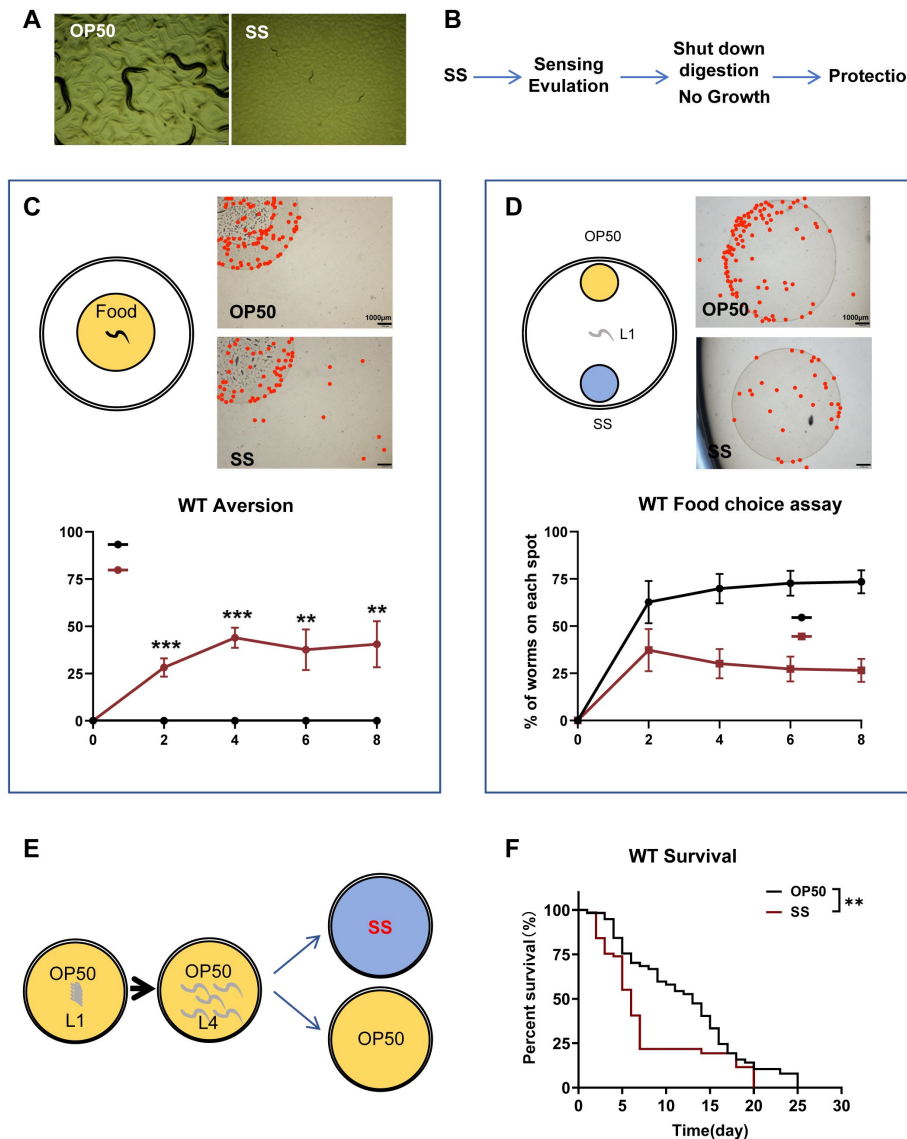


Figure 1.

***C. elegans* shuts down its digestion for survival when fed harmful food, specifically *Staphylococcus saprophyticus* (SS).**

(A) Microscopic images showing worms fed with SS arrested at the L1 stage three days after hatching.

(B) Schematic model illustrating our hypothesis: *C. elegans* can sense or evaluate inedible food, such as SS, and subsequently shut down their digestion to arrest development as a protective survival strategy.

(C) Schematic drawing and quantitative data of the food dwelling/avoidance assay. Yellow circles indicate the food spot for OP50 or SS bacteria, respectively. The animals were scored at the indicated times after L1 worms were placed on the food spot. The red point indicates the position of each worm. Data are represented as mean $\pm$ SD. Scale bar = 1000 μ m. ***p < 0.001; **p < 0.01 by Student's t-test.

(D) Schematic drawing, microscopic images, and quantitative data of the food choice assay. L1 worms were placed at the center spot (origin). OP50 (yellow) and SS (blue) bacteria were placed on opposite sides of the plate. The red point indicates the position of each worm. The percentage of worms on each spot was calculated at the indicated times. Data are represented as mean $\pm$ SD. Scale bar = 1000 μ m. ****p < 0.0001; **p < 0.01 by Student's t-test.

(E-F) Schematic drawing and quantitative data of the lifespan of animals fed with SS or OP50. L1 worms were seeded onto OP50 and grown to the L4 stage. L4 worms were then moved to SS or OP50 food to measure lifespan. **p < 0.01 by log-rank test. All data are representative of at least three independent experiments.

that cannot sense SS, thereby allowing its digestion and supporting growth. One of the mutant alleles identified, *ylf6*, could digest SS and exhibited a growth phenotype under SS feeding conditions (Figure 2—figure supplement 1B). Whole-genome deep sequencing revealed that *ylf6* carries two mutations in the *nsy-1* gene (H929Y, Q1191^[stop codon*]) (Figure 2—figure supplement 1C).

To confirm that *nsy-1* is essential for shutting down SS digestion, we used an independent *nsy-1* mutant allele, *ag3*, and found that *nsy-1(ag3)* mutants also digested SS (Figure 2A). To determine whether *nsy-1* is crucial for sensing SS, we performed two behavioral assays: food dwelling/avoidance and food choice. In the food dwelling/avoidance assay, larval stage *nsy-1* mutant exhibited significantly impaired avoidance responses at both 4h and 6h but not at 8h, suggesting that NSY-1 is essential for sustained aversion to SS food in the early response (Figure 2B). In contrast to wild-type N2 animals, *nsy-1* mutants preferred SS when given a choice between two poor-quality foods, heat-killed *E. coli* and SS (Figure 2C, Figure 1—figure supplement 1B). These results suggest that *nsy-1* is essential for *C. elegans* to sense and avoid SS.

Next, we examined whether larvae that cannot sense SS and do not shut down digestion can adapt to SS environments. We measured the survival rate of *nsy-1* mutants under SS feeding conditions and found that these mutants had a higher mortality rate (Figure 2D).

Overall, these results indicate that *C. elegans* sense and detect unfavorable food, such as SS, through *nsy-1*, and subsequently shut down digestion to protect themselves and enhance survival.

NSY-1 Functions in AWC Neurons to Shut Down SS Digestion

The *nsy-1* gene in *C. elegans* encodes a MAP kinase kinase kinase (MAPKKK) that operates in the AWC neurons (Kim et al., 2002; Sagasti et al., 2001), which are essential for chemotaxis and odor sensation. The primary role of *nsy-1* in AWC neurons is to regulate the asymmetric expression of odorant receptors, contributing to neuronal asymmetry and diversity (Chuang et al., 2007; Sagasti et al., 2001). We hypothesized that the shutdown of SS digestion in *C. elegans* is mediated by *nsy-1* function in AWC neurons.

Firstly, we constructed a *Pnsy-1::GFP* reporter strain and confirmed that *nsy-1* is expressed in the AWC neurons (Figure 3A). Secondly, we expressed *nsy-1* in the AWC neurons of *nsy-1* mutant animals and observed that the transgenic animals rescued the indigestion phenotype (Figure 3B). This indicates that *nsy-1* functions in AWC neurons to shut down SS digestion. Thirdly, we used CRISPR to construct a mutant strain that knocks out *nsy-1* specifically in AWC neurons (Figure 3—figure supplement 1). We found that *nsy-1* knockout in AWC neurons also resulted in the shutdown of SS digestion (Figure 3C). These results collectively suggest that NSY-1 is functional in AWC neurons and is crucial for shutting down SS digestion in *C. elegans*.

Beyond its established role in AWC neurons, we detected NSY-1 expression in the intestine (Figure 3-figure supplement 2A). To assess intestinal NSY-1 function, we performed tissue-specific rescue experiments in *nsy-1* mutants using the intestinal-specific *vha-6* promoter. Intestinal expression of NSY-1 significantly suppressed the enhanced SS digestion phenotype in *nsy-1* mutants (Figure 3-figure supplement 2B), demonstrating functional involvement of gut-localized NSY-1 in regulating digestive responses. We propose intestinal NSY-1 mediates this effect through innate immune signaling, consistent with its known pathway components. As our previously established (Geng et al., 2022), the canonical PMK-1/p38 MAPK pathway functions downstream of NSY-1, with both *sek-1* and *pmk-1* knockdown enhancing SS digestion through immune modulation. This indicates intestinal NSY-1 suppresses digestion may act through PMK-1-mediated immune responses. Since neuronal NSY-1's role in digestive control was previously undefined, we prioritized mechanistic analysis of its neuronal function in digestion regulation.

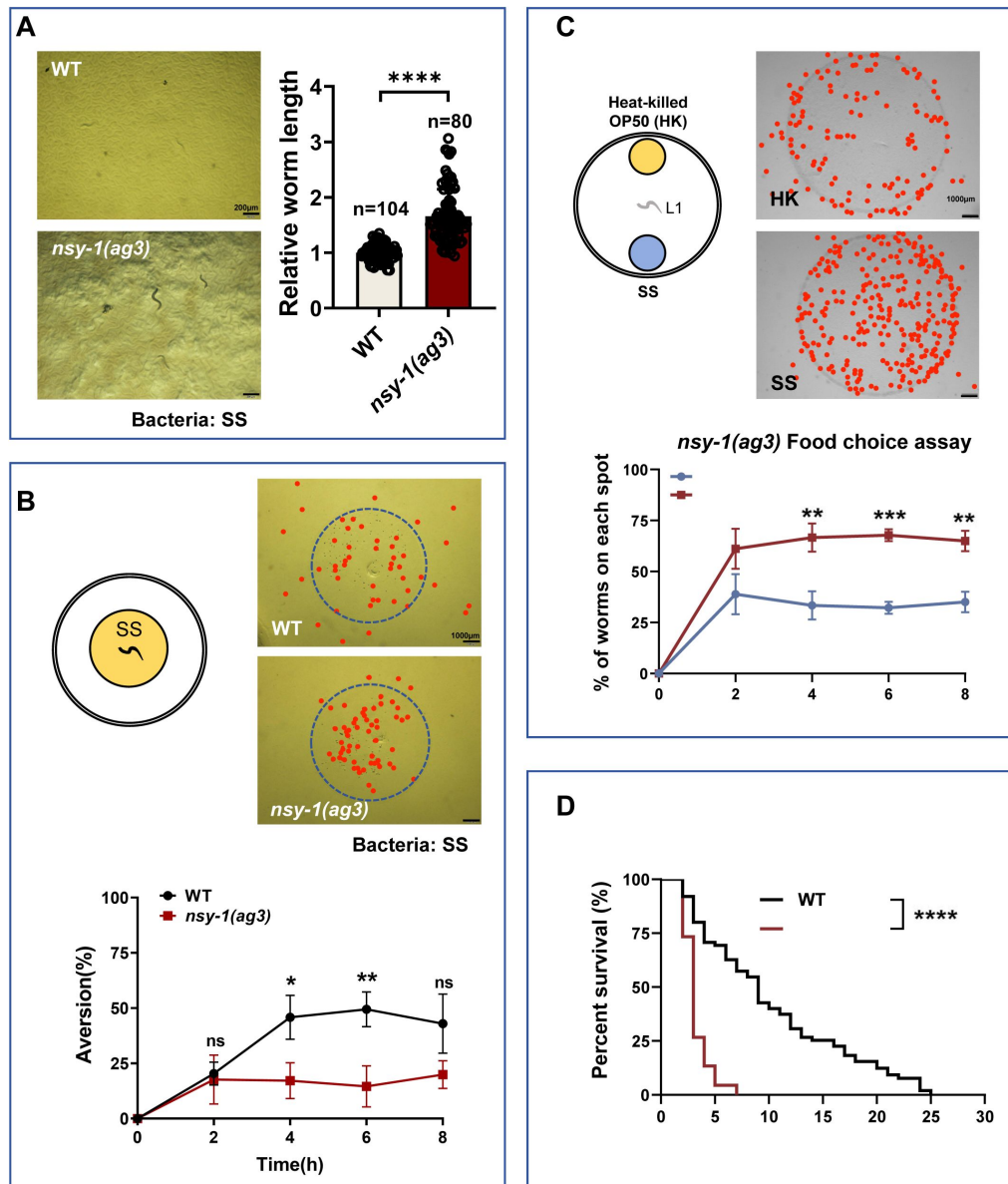


Figure 2.

C. elegans senses SS and shuts down digestion through NSY-1.

(A) Developmental phenotype of wild-type N2 and *nsy-1(ag3)* mutant worms fed with SS bacteria. Data are represented as mean $\pm$ SD. Scale bar = 200 μ m. **** p < 0.0001 by Student's *t*-test. *n* = number of animals which were scored.

(B) Schematic drawing, microscopic images, and quantitative data of the food dwelling/avoidance assay. Yellow circles indicate the food spot for SS bacteria. The animals were scored at the indicated times after L1 worms were placed on the food spot. The blue circle indicates the edge of the bacterial lawn, and the red point indicates the position of each worm. Data are represented as mean $\pm$ SD. Scale bar = 1000 μ m. * p < 0.05; ** p < 0.01 by Student's *t*-test.

(C) Schematic drawing, microscopic images, and quantitative data of the food choice assay. L1 *nsy-1(ag3)* worms were placed at the center spot (origin). Heat-killed OP50 (yellow) and SS (blue) bacteria were placed on opposite sides of the plate. The red point indicates the position of each worm. The percentage of worms on each spot was calculated at the indicated times. Data are represented as mean $\pm$ SD. Scale bar = 1000 μ m. ** p < 0.01; *** p < 0.001 by Student's *t*-test.

(D) Survival curves of wild-type N2 and *nsy-1(ag3)* mutant worms fed with SS bacteria. L4 worms, previously fed OP50 bacteria, were transferred to SS food to measure lifespan. **** p < 0.0001 by log-rank test.

All data are representative of at least three independent experiments.

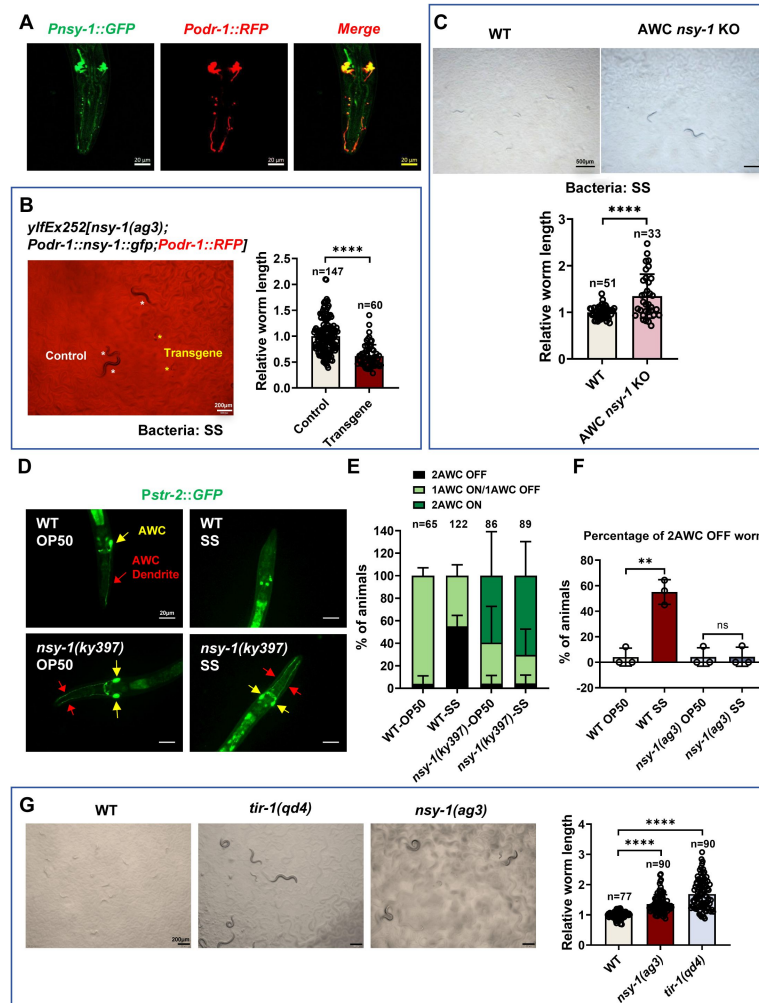


Figure 3.

NSY-1 plays a critical role in AWC neurons to inhibit SS digestion.

(A) Microscopic image showing the expression pattern of *nsy-1*. The head of an adult transgenic animal carrying *Pnsy-1::GFP* and *Podr-1::RFP* shows colocalization of *nsy-1* and *odr-1*. Scale bar = 20 μ m.

(B) Developmental progression of *nsy-1(ag3)* mutant worms carrying *Podr-1::nsy-1::gfp* (AWC neuron-specific expression) grown on SS bacteria. Control animals are labeled with white stars, and animals carrying the transgenes (rescued animals) are labeled with yellow stars. Data are represented as mean $\pm$ SD. Scale bar = 200 μ m. **** p < 0.0001 by Student's *t*-test. *n* = number of animals which were scored.

(C) Developmental progression of wild-type N2 and AWC neuron-specific knockout *nsy-1* animals (AWC *nsy-1* KO) grown on SS bacteria. Data are represented as mean $\pm$ SD. Scale bar = 500 μ m. *** p < 0.001 by Student's *t*-test. *n* = number of animals which were scored.

(D) Microscopic images show *Pstr-2::GFP*, a marker for AWC neuron states, in L1-staged wild-type and *nsy-1(ky397)* mutant worms grown on OP50 or SS bacteria for 6 hours. AWC neuron positions are highlighted with red and yellow arrows. Scale bar = 20 μ m.

(E-F) Percentage of animals with different AWC neuron states. *nsy-1* mutation promotes an 2AWC^{ON} state under SS feeding conditions (E), with approximately 50% of animals exhibiting 2AWC^{OFF} neurons when feeding on SS (F). Data are represented as mean $\pm$ SD. ** p < 0.01 by Student's *t*-test. *n* = number of animals which were scored.

(G) Developmental progression of wild-type N2, *tir-1(qd4)*, and *nsy-1(ag3)* mutant worms grown on SS bacteria. Data are represented as mean $\pm$ SD. Scale bar = 200 μ m.

**** p < 0.0001 by Student's *t*-test. *n* = number of animals which were scored. All data are representative of at least three independent experiments.

To determine whether NSY-1 in AWC neurons mediates SS sensory perception, we performed dwelling (avoidance) and food-choice assays using AWC-specific *nsy-1* knockout and AWC-rescued strains (*nsy-1(ag3)*; *Podr-1::nsy-1*). In dwelling assays, AWC-specific *nsy-1* KO mutants exhibited significantly impaired SS avoidance at 6h (Figure 3-figure supplement 3A), while AWC-rescued strains restored avoidance capacity at 2-6h (Figure 3-figure supplement 3B). Food-choice assays further revealed that AWC *nsy-1* KO mutants preferentially migrated toward SS (Figure 3-figure supplement 3C), whereas AWC-rescued showed no preference between SS and HK-*E. coli* (Figure 3-figure supplement 3D). These data conclusively demonstrate that NSY-1 acts in AWC neurons to mediate SS recognition and aversion behaviors.

AWC Neurons Exhibit OFF State in Sensing SS Food

In *C. elegans*, the expression of the *str-2* gene in AWC neurons indicates the ON state (AWC^{ON}) (Sagasti et al., 2001 [\[4\]](#)), which is associated with high cGMP levels and lower calcium activity, enabling the neuron to respond to specific odors. Conversely, the absence of *str-2* expression marks the OFF state (AWC^{OFF}), characterized by different odor responses, low cGMP levels, and higher calcium activity (Troemel et al., 1999 [\[4\]](#)). We aimed to investigate (1) whether AWC neurons exhibit different states under normal food (*E. coli* OP50) versus unfavorable food (SS) conditions and (2) whether the state of AWC neurons affects the ability of *C. elegans* to digest SS.

Using *str-2::GFP* as a marker for AWC neuron states (Troemel et al., 1999 [\[4\]](#)), we found that wild-type animals feeding on normal OP50 food typically exhibit one AWC^{OFF} and one AWC^{ON} neuron. However, when feeding on SS, the proportion of animals with the AWC^{OFF} state increased, with approximately 50% of animals exhibiting two AWC^{OFF} neurons (Figure 3D, E, F [\[4\]](#)).

In *nsy-1* mutant animals feeding on OP50, *str-2::GFP* is expressed in both AWC neurons (2AWC^{ON}), consistent with previous studies (Chuang and Bargmann, 2005 [\[4\]](#); Chuang et al., 2007 [\[4\]](#); Sagasti et al., 2001 [\[4\]](#); Troemel et al., 1999 [\[4\]](#)) (Figure 3D, E [\[4\]](#)). Notably, both AWC neurons remained in the AWC^{ON} state in *nsy-1* mutants feeding on SS (Figure 3D, E [\[4\]](#)). These results suggest that SS feeding induces an AWC^{OFF} state in wild-type animals, while the *nsy-1* mutation promotes an AWC^{ON} state even under SS feeding conditions. This implies that the AWC^{OFF} state may inhibit SS digestion, whereas the AWC^{ON} state promotes it.

The TIR-1-NSY-1-SEK-1-MAPK pathway plays a crucial role in regulating the asymmetric AWC cell fate decision. Previous studies have shown that *str-2::GFP* expression in both AWC cells (2AWC^{ON} phenotype) occurs in *tir-1*, *nsy-1*, and *sek-1* mutant animals (Chuang and Bargmann, 2005 [\[4\]](#); Sagasti et al., 2001 [\[4\]](#); Troemel et al., 1999 [\[4\]](#)). We found that *tir-1* mutant can grow under SS feeding conditions (Figure 3G [\[4\]](#)), indicating that *tir-1* mutant can digest SS similarly to *nsy-1* mutants. This demonstrates that the AWC^{ON} state promotes SS digestion.

To confirm the important of AWC state in SS digestion, we performed AWC-specific neuron ablation experiments using previously validated transgenic strain that expresses cleaved caspase under the AWC-specific promoter, *ceh-36* (*ceh-36p::caspase*). Critically, worms with ablated AWC neurons completely failed to digest SS food (Figure 3—figure supplement 4), phenocopying the non-digesting state of wild-type worms on SS. This result directly confirms that functional AWC neurons are essential for initiating SS digestion, aligning with our model where the AWC-OFF state (induced by SS) inhibits digestion while the AWC-ON state promotes it.

Overall, our data suggest that the state of AWC neurons is critical for sensing food in *C. elegans*. When sensing SS food, animals exhibit an AWC^{OFF} state, which shuts down digestion. Conversely, the AWC^{ON} state promotes the digestion of SS.

NSY-1 Shuts Down SS Digestion through Induction of STR-130

We have demonstrated that NSY-1 in AWC neurons detects unfavorable food and shuts down digestion (**Figure 3**). To investigate the genes regulated by NSY-1 in response to SS and their impact on SS digestion, we conducted a transcriptomic analysis on L1 larval animals fed with normal food (OP50) or unfavorable food (SS) for a short duration (4 hours). We speculated that some genes induced by SS food are dependent on NSY-1 (**Figure 4A**), and their induction aids in shutting down SS digestion.

RNA-seq data analysis revealed 304 NSY-1-dependent candidate genes responding to SS (**Figure 4B**, Supplementary file 1). Enrichment analysis (Figure 4—figure supplement 1A, Supplementary file 2) of these candidate genes showed mainly associations with biotic stimulus, defense responses, xenobiotic stimulus, suggesting that NSY-1 positively regulates stress response pathways to protect animals under harmful food, SS, feeding conditions. Moreover, we found that sensory perception related genes (*sra-32*, *str-87*, *str-112*, *str-130*, *str-160*, *str-230*) (Figure 4—figure supplement 1A, Supplementary file 2) were also enriched, with many genes being G protein-coupled receptors (GPCRs), which mediate odor sensing (Buck and Axel, 1991).

We further analyzed the dependence of these enriched GPCRs on NSY-1 under SS feeding conditions (Figure 4—figure supplement 1B) and found that *str-130* is significantly upregulated in response to SS, with its function strong being NSY-1 dependent (**Figure 4C**).

Using RNAi knockdown and the SS growth assay, we observed that RNAi of *str-130*, *str-230*, *str-87*, or *str-112* significantly enhanced SS growth (Figure 4—figure supplement 2A), with *str-130* RNAi exhibiting the most robust phenotype—phenocopying *nsy-1(ag3)* mutants. Crucially, none of these GPCR knockdowns further enhanced growth in *nsy-1(ag3)* mutants (Figure 4—figure supplement 2B), confirming their position downstream of NSY-1. These data establish *str-130* as the dominant effector of NSY-1-mediated SS response regulation, while suggesting minor contributions from other GPCRs (*str-230*, *str-87*, *str-112*).

It has been shown that *str-130* is expressed in AWC^{OFF} neurons, based on transgenic GFP reporter strains, *str-130p::GFP* (Vidal et al., 2018). Our data also show that *str-130* expression is induced in wild-type animals fed with SS (**Figure 4C**), where AWC neurons exhibit the AWC^{OFF} state (**Figure 3D, E**). Therefore, it is possible that the high expression of *str-130*, regulated by *nsy-1*, alters the AWC state and inhibits SS digestion.

Firstly, we found that knockdown of *str-130* in wild-type animals promoted SS digestion, thereby supporting animal growth (**Figure 4D**), and the proportion of animals with two AWC^{OFF} neurons decreased (**Figure 4E**). Secondly, we found that overexpression of *str-130* in *nsy-1* mutant animals inhibited SS digestion, thereby slowing animal growth (**Figure 4F**), and the proportion of animals with two AWC^{OFF} neurons increased (**Figure 4G**). These results demonstrate that NSY-1 promotes the AWC^{OFF} state by inducing *str-130* expression, which in turn inhibits SS digestion in *C. elegans*.

To definitively establish the epistatic relationship between NSY-1 and STR-130, we performed RNAi knockdown of *str-130* in the *nsy-1(ag3)* mutant background and assessed development on SS food. We found that the *str-130* RNAi did not further enhance the developmental capacity of *nsy-1(ag3)* mutant animals on SS (Figure 4—figure supplement 3). This epistasis confirms STR-130 functions strictly downstream of NSY-1 within the same genetic pathway. Together with our overexpression data (**Figure 4F-G**) showing neuronal *str-130* rescue suppresses SS digestion in *nsy-1* mutants, these results establish a linear signaling axis where NSY-1 primarily achieves functional inhibition of SS digestion through induction of the GPCR *str-130*.

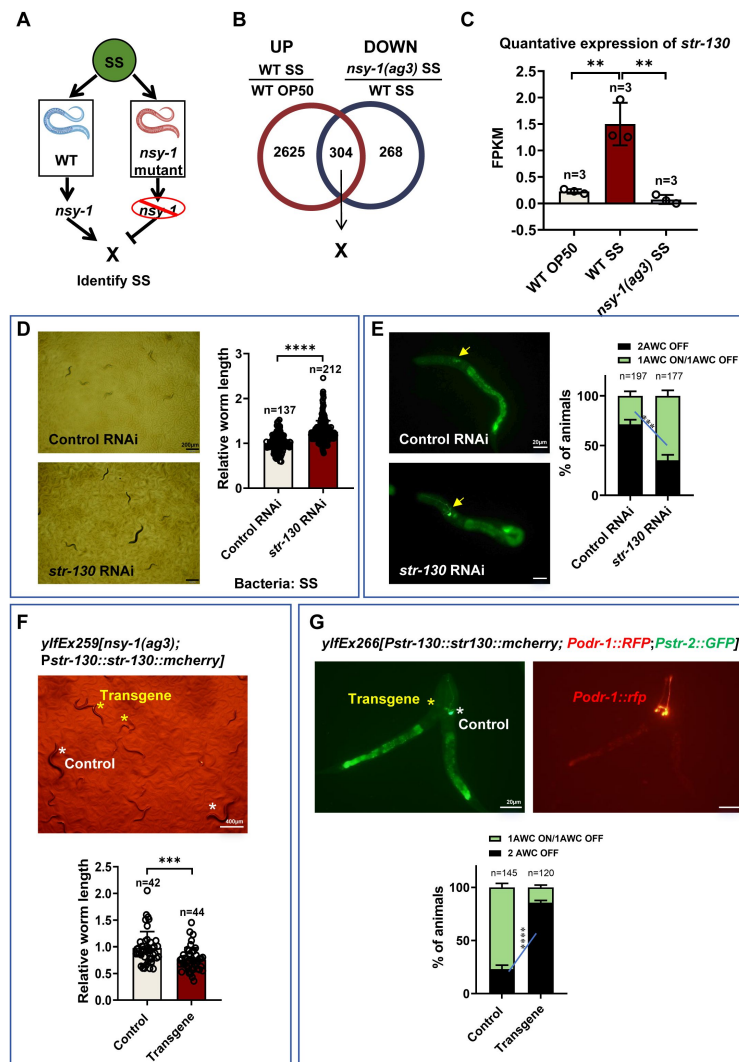


Figure 4.

NSY-1 inhibits animals from digesting SS by inducing *str-130*.

(A) Schematic illustration showing that “X” genes rely on NSY-1 to shut down SS digestion. “X” genes induced by SS food are dependent on NSY-1, and their induction aids in shutting down SS digestion.

(B) Venn diagram showing the overlap of genes that respond to SS and rely on NSY-1. The number of genes is indicated in the diagram (also see Supplementary file 1).

(C) Transcriptome analysis showing *str-130* mRNA expression, which relies on NSY-1 in response to SS. Data are represented as mean $\pm$ SD. ** $p < 0.01$ by Student’s t-test.

(D) Developmental progression of wild-type animals treated with control RNAi or *str-130* RNAi grown on SS bacteria. Data are represented as mean $\pm$ SD. Scale bar= 200 μ m. **** $p < 0.0001$ by Student’s t-test. n= number of animals which were scored.

(E) Microscopic images and quantitative data of AWC neuron states in L1 animals treated with control RNAi or *str-130* RNAi grown on SS bacteria. Data are represented as mean $\pm$ SD. Scale bar= 20 μ m. *** $p < 0.001$ by Student’s t-test (1AWC^{ON}/1AWC^{OFF}: control vs *str-130* RNAi). n= number of animals which were scored.

(F) Developmental progression of *nsy-1(ag3)* mutant worms carrying *Pstr-130::str-130::mCherry* grown on SS bacteria. Control animals are labeled with white stars, and animals carrying transgenes are labeled with yellow stars. Data are represented as mean $\pm$ SD. Scale bar= 400 μ m. *** $p < 0.001$ by Student’s t-test. n= number of animals which were scored.

(G) Microscopic images and quantitative data of AWC neuron states in L1 animals carrying *Pstr-130::str-130::mCherry*. Transgenic animals with overexpression of *str-130* (carrying *Pod-1::GFP* as a co-injection marker) show an increased 2AWC^{OFF} state. Data are represented as mean $\pm$ SD. Scale bar= 20 μ m. *** $p < 0.001$ by Student’s t-test (2AWC^{OFF}: Control vs Transgene). n= number of animals which were scored.

All data are representative of at least three independent experiments.

NSY-1 Mutation Promotes SS Digestion by Inducing Insulin Signaling

NSY-1 mutation promotes the digestion of unfavorable food, such as SS, and supports *C. elegans* growth (Figure 2A). We hypothesize that, in addition to upregulating certain genes, such as *str-130*, to inhibit SS digestion, NSY-1 may also suppress certain genes to prevent nematodes from utilizing SS (Figure 5A). One possibility is that some genes, induced by the *nsy-1* mutation under SS feeding conditions could facilitate SS digestion in the *nsy-1* mutant (Figure 5A).

Our RNA-seq analysis identified 308+46=354 genes that are induced by the *nsy-1* mutation under SS feeding conditions (Figure 5B, Supplementary file 3). However, among these 354 genes, 46 genes can also be induced in wild-type animals fed with SS, suggesting that these 46 genes may not be involved in digesting SS in *nsy-1* mutants. Therefore, the 308 candidate genes induced by the *nsy-1* mutation under SS feeding conditions could potentially promote animals to digest SS.

Enrichment analysis revealed that genes related to extracellular functions, such as insulin-related genes, are induced in *nsy-1* mutant animals (Figure 5—figure supplement 1A, Supplementary file 4). Further analysis of insulin-related genes from the RNA-seq data showed that *ins-23* is predominantly induced in *nsy-1* mutant animals (Figure 5—figure supplement 1B), suggesting its potential role in promoting SS digestion.

To determine if insulin-like peptide genes were functionally responsible for the enhanced SS growth observed in *nsy-1(ag3)* mutants, we performed functional phenotypic screening using the SS growth assay (worm length assay). We individually knocked down each of these candidates (*ins-22*, *ins-23*, *ins-24*, *ins-27*) in the *nsy-1(ag3)* mutant background. Among these, only RNAi targeting *ins-23* significantly suppressed the enhanced development of the *nsy-1(ag3)* mutant on SS (Figure 5—figure supplement 2, Figure 5D). This targeted functional screening revealed that *ins-23* has the most robust and specific role in mediating the enhanced digestion phenotype downstream of NSY-1 loss, providing the critical justification for our subsequent focus on this particular insulin-like peptide.

Given that INS-23 is expressed in AWC neurons (Figure 5—figure supplement 3A, from CeNGEN), this suggests increased production and likely enhanced release of INS-23 from AWC neurons in the *nsy-1(ag3)* mutant background, which promotes SS digestion.

The insulin/insulin-like growth factor signaling (IIS) pathway, particularly through the DAF-2 receptor, integrates nutritional signals to regulate various behavioral and physiological responses related to food (Kodama et al., 2006; Ryu et al., 2018). It has been shown that INS-23 acts as an antagonist for the DAF-2 receptor to promote larval diapause (Matsunaga et al., 2018). To test whether *ins-23* induction in *nsy-1(ag3)* mutants promotes SS digestion through its receptor, DAF-2, we constructed a *nsy-1; daf-2* double mutant. We found that the SS digestion ability of the *nsy-1* mutant was inhibited by the *daf-2* mutation (Figure 5E). This suggests that the *nsy-1* mutation induces the insulin peptide *ins-23*, which promotes SS digestion through its potential receptor, DAF-2.

To investigate whether DAF-2 acts as the gut-localized receptor for neuronal INS-23 signaling, we performed tissue-specific rescue experiments in the *nsy-1(ag3); daf-2(e1370)* double mutant. When DAF-2 was re-introduced specifically in the intestine (using the *ges-1* promoter), we observed a significant suppression of SS digestion (Figure 5—figure supplement 3B), but not rescue digestive defect. This indicates that INS-23 induction in *nsy-1* mutants promotes digestion independently of intestinal DAF-2 function.

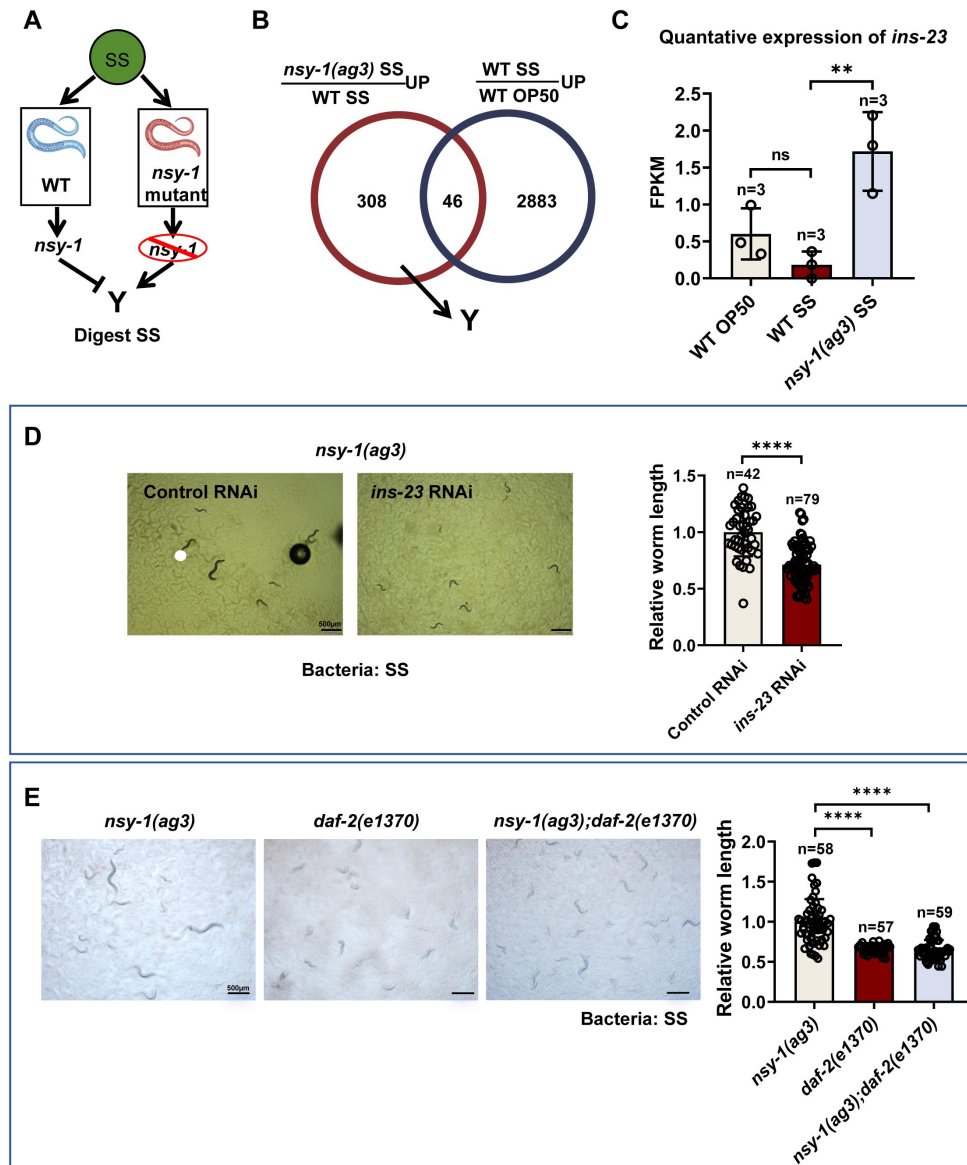


Figure 5.

NSY-1 mutation activates animals to digest SS by inducing insulin signaling.

(A) Schematic illustration showing that NSY-1 inhibits the expression of "Y" genes, which promote SS digestion. Some genes induced by the *nsy-1* mutation under SS feeding conditions could facilitate SS digestion in the *nsy-1* mutant.

(B) Venn diagram showing the overlap of genes that respond to SS but are limited by NSY-1. A total of 308 candidate genes induced by the *nsy-1* mutation under SS feeding conditions could potentially promote SS digestion.

(C) Transcriptome analysis showing that *ins-23* expression is induced in animals with the *nsy-1* mutation under SS feeding conditions. Data are represented as mean $\pm$ SD.

**p < 0.01; n.s. not significant by Student's t-test. n=three biological replicates.

(D) Developmental progression of *nsy-1(ag3)* mutant animals treated with control RNAi or *ins-23* RNAi grown on SS bacteria. Data are represented as mean $\pm$ SD. Scale bar = 500 μ m. ****p < 0.0001 by Student's t-test. n= number of animals which were scored.

(E) Developmental progression of *nsy-1(ag3)*, *daf-2(e1370)* and *nsy-1(ag3);daf-2(e1370)* double mutant animals grown on SS bacteria. Data are represented as mean $\pm$ SD. Scale bar = 500 μ m. ****p < 0.0001 by Student's t-test. n= number of animals which were scored.

All data are representative of at least three independent experiments.

As established in our prior work (Geng et al., 2022 [DOI](#)), SS exposure triggers phosphorylation of PMK-1 (P-PMK-1) in *C. elegans*, and *pmk-1* mutants exhibit enhanced growth on SS. This confirms that PMK-1-mediated innate immune signaling actively regulates SS responsiveness and digestion. To address whether PMK-1 functions downstream of NSY-1 within our proposed model, we performed critical epistasis analyses. While we observed that *nsy-1* mutation elevates *ins-23* (indicating NSY-1 suppression of *ins-23*), knockdown of *pmk-1* did not alter *ins-23* expression levels (Figure 5—figure supplement 3C). This demonstrates that PMK-1 does not operate through the INS-23 pathway to regulate SS digestion. Thus, although both pathways respond to SS, the PMK-1-mediated innate immune response and the NSY-1/INS-23 axis constitute distinct regulatory mechanisms governing digestive adaptation.

NSY-1 Mutation Promotes SS Digestion through Regulation of Intestinal BCF-1

In our previous study, we found that heat-killed *E. coli* promotes SS digestion in *C. elegans* (Geng et al., 2022 [DOI](#)), a process requiring intestinal BCF-1 (Hao et al., 2024 [DOI](#)). In the absence of BCF-1, the digestive capability of the animals is significantly reduced (Hao et al., 2024 [DOI](#)). This led us to investigate whether NSY-1 in AWC neurons regulates intestinal *bcf-1* expression.

Firstly, we used a BCF-1::GFP reporter to measure *bcf-1* expression in *nsy-1* mutant animals. We found that mutation of *nsy-1* induced *bcf-1* expression in animals fed either SS or OP50 food (Figure 6A [DOI](#)). Additionally, we confirmed that *nsy-1* mutation in AWC neurons also induced intestinal *bcf-1* expression under SS feeding conditions (Figure 6B [DOI](#)). This data indicates that NSY-1 in AWC neurons inhibits intestinal *bcf-1* expression, implying that *nsy-1* mutation promotes SS digestion through BCF-1.

Next, we constructed a *nsy-1; bcf-1* double mutant and analyzed the growth of these animals on SS. We found that the digestive ability of *nsy-1* mutants was inhibited by the mutation of *bcf-1* (Figure 6C [DOI](#)). Together, our data suggest that the increased digestion ability in *nsy-1* mutant animals is dependent on the intestinal digestion factor BCF-1.

We then asked how *nsy-1* regulates intestinal *bcf-1* expression. Since *nsy-1* mutation induces the insulin peptide *ins-23*, which promotes SS digestion, we tested whether the induction of intestinal *bcf-1* by *nsy-1* mutation is also mediated through INS-23. We found that the *bcf-1*::GFP level decreased in *nsy-1* mutant animals following *ins-23* RNAi treatment (Figure 6D [DOI](#)). This suggests that *nsy-1* mutation activates *bcf-1* expression in the intestine, which requires INS-23.

To test whether INS-23 acts in AWC neurons to regulate intestinal BCF-1, we generated AWC-specific knockdown strains which was achieved by rescuing *sid-1* cDNA expression under the *ceh-36* promoter in a *sid-1(qt9);BCF-1::GFP* background. We found that AWC-restricted *ins-23* knockdown significantly reduced intestinal BCF-1::GFP expression (Figure 6—figure supplement 1A). This confirms that INS-23 functions within AWC sensory neurons to activate intestinal BCF-1, consistent with NSY-1's upstream inhibition of INS-23 in this neuronal subtype.

NSY-1 promotes the AWC^{OFF} state through STR-130 to suppress SS digestion. To determine whether AWC-expressed STR-130 regulates intestinal BCF-1 expression, We observed that AWC neuron-specific RNAi of *str-130* elevated intestinal BCF-1::GFP expression (Figure 6—figure supplement 1B). This demonstrates that STR-130 functions in AWC neurons to repress BCF-1 expression in the intestine.

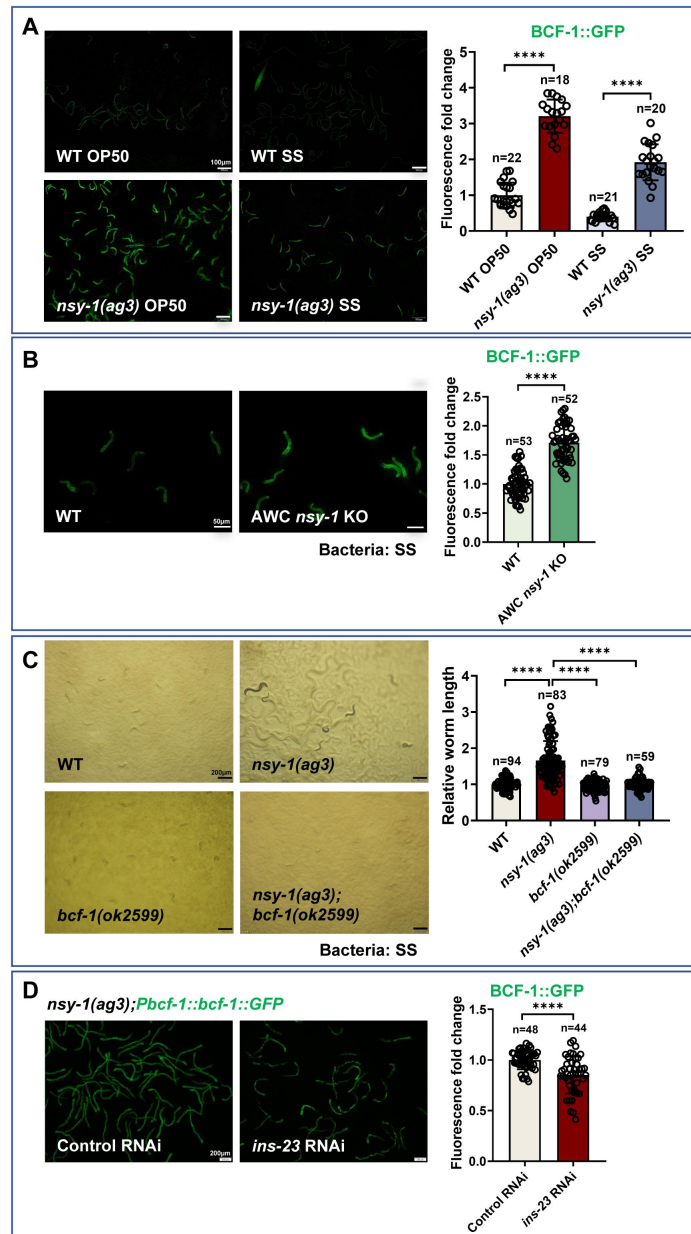


Figure 6.

NSY-1 mutation promotes animals to digest SS through inducing intestinal *bcf-1*.

(A) Microscopic images and quantitative data showing fluorescence of *Pbcf-1::bcf-1::GFP* in L1-staged wild-type (WT) and *nsy-1(ag3)* mutant animals fed with OP50 or SS bacteria for 6 hours. Data are represented as mean $\pm$ SD. Scale bar = 100 μ m.

**** $p < 0.0001$ by Student's t-test. n = number of animals which were scored.

(B) Microscopic images and quantitative data showing fluorescence of *Pbcf-1::bcf-1::GFP* in L1-staged wild-type and AWC *nsy-1* KO mutant (AWC neuron-specific knockout *nsy-1* animals) fed with SS bacteria for 6 hours. Data are represented as mean $\pm$ SD. Scale bar = 50 μ m. **** $p < 0.0001$ by Student's t-test. n = number of animals which were scored.

(C) Developmental progression of wild-type N2, *nsy-1(ag3)*, *bcf-1(ok2599)*, and *nsy-1(ag3); bcf-1(ok2599)* double mutant animals grown on SS bacteria. Data are represented as mean $\pm$ SD. Scale bar = 200 μ m. **** $p < 0.0001$ by Student's t-test. n = number of animals which were scored.

(D) Microscopic images and quantitative data showing fluorescence of *Pbcf-1::bcf-1::GFP* in *nsy-1(ag3)* mutant animals treated with control RNAi or *ins-23* RNAi under normal RNAi feeding conditions. Data are represented as mean $\pm$ SD. Scale bar = 200 μ m. **** $p < 0.0001$ by Student's t-test. n = number of animals which were scored.

All data are representative of at least three independent experiments.

Discussion

This study in *Caenorhabditis elegans* reveals a neural-digestive mechanism for evaluating harmful food (**Figure 7**). The neuron-expressed NSY-1 protein detects *Staphylococcus saprophyticus* (SS) as unsafe food, triggering a digestive shutdown via the AWC^{OFF} neural circuit and the NSY-1-dependent STR-130. Mutations in NSY-1 lead to SS digestion, activating the insulin/IGF-1 signaling (IIS) pathway and BCF-1 expression. These findings highlight a food quality evaluation strategy where neurons communicate with the digestive system to assess food safety, providing insights into how animals adapt to toxic food environments.

The identification of NSY-1 in AWC neurons as a key player in detecting *Staphylococcus saprophyticus* (SS) and initiating digestive shutdown is particularly intriguing. NSY-1, a MAP kinase (MAPKKK), is known to regulate the asymmetric expression of odorant receptors, contributing to neuronal asymmetry and diversity (Chuang and Bargmann, 2005; Chuang et al., 2007; Sagasti et al., 2001; Troemel et al., 1999). Previous research has demonstrated that AWC neurons are pivotal in chemotaxis and sensory processing (Bargmann et al., 1993; Troemel et al., 1997; Wes and Bargmann, 2001). Our findings extend its function to include food quality assessment, showing that NSY-1 can trigger a digestive shutdown via the AWC^{OFF} neural circuit.

This neural circuit appears to be a crucial component of a systemic food quality control mechanism, allowing animals to adapt effectively to harmful food sources. The state of AWC neurons (AWC^{ON} or AWC^{OFF}) directly influences the animal's ability to digest SS, with the AWC^{OFF} state inhibiting digestion and the AWC^{ON} state promoting it. The finding that activation of the AWC^{OFF} neural circuit leads to a systemic digestive shutdown mediated by NSY-1-dependent GPCR(STR-130) is another notable advancement. GPCRs are well-known for their roles in sensory perception (Julius and Nathans, 2012; Troemel et al., 1995). Our results suggested that GPCR STR-130 play a role in shutting down digestive processes through maintaining AWC^{OFF} states for evaluating harmful food.

Our study underscores the critical role of insulin signaling pathways in mediating the effects of neuronal detection on intestinal functions. Upon detection of SS by AWC neurons, NSY-1 inhibits the expression of insulin-like peptides, particularly INS-23. These neuronal peptides promote the expression of BCF-1, a key regulatory factor in digestion (Hao et al., 2024). Once *ins-23* was inhibited by neuronal NSY-1, the intestinal BCF-1 level is also reduced, which in turn shuts down digestion. We found that the digestive ability of *nsy-1* mutants was totally inhibited by the mutation of *bcf-1* (**Figure 6C**), suggesting that the increased digestion ability in *nsy-1* mutant animals is mainly dependent on the intestinal digestion factor BCF-1. This regulatory cascade highlights the intricate link between neuronal signals and gut responses, ensuring an adaptive reaction to harmful food. We speculated that except INS-23, there should be other factors as signaling regulated by neuronal NSY-1 to inhibit intestinal digestion factor BCF-1 for digestion shutdown.

Future studies should focus on delineating the precise molecular pathways linking NSY-1 signaling in AWC neurons to digestion regulation in the gut. Identifying the role of other sensory neurons in food quality assessment and their interactions with the digestive system may uncover new facets of neuron-gut communication and adaptive responses.

In summary, our study reveals a sophisticated mechanism in *C. elegans* that integrates neuronal detection of harmful food sources with systemic digestive responses. This neuron-digestive crosstalk is crucial for maintaining organismal homeostasis and survival in the presence of toxic food sources. The findings suggest that similar pathways may exist in other species, including

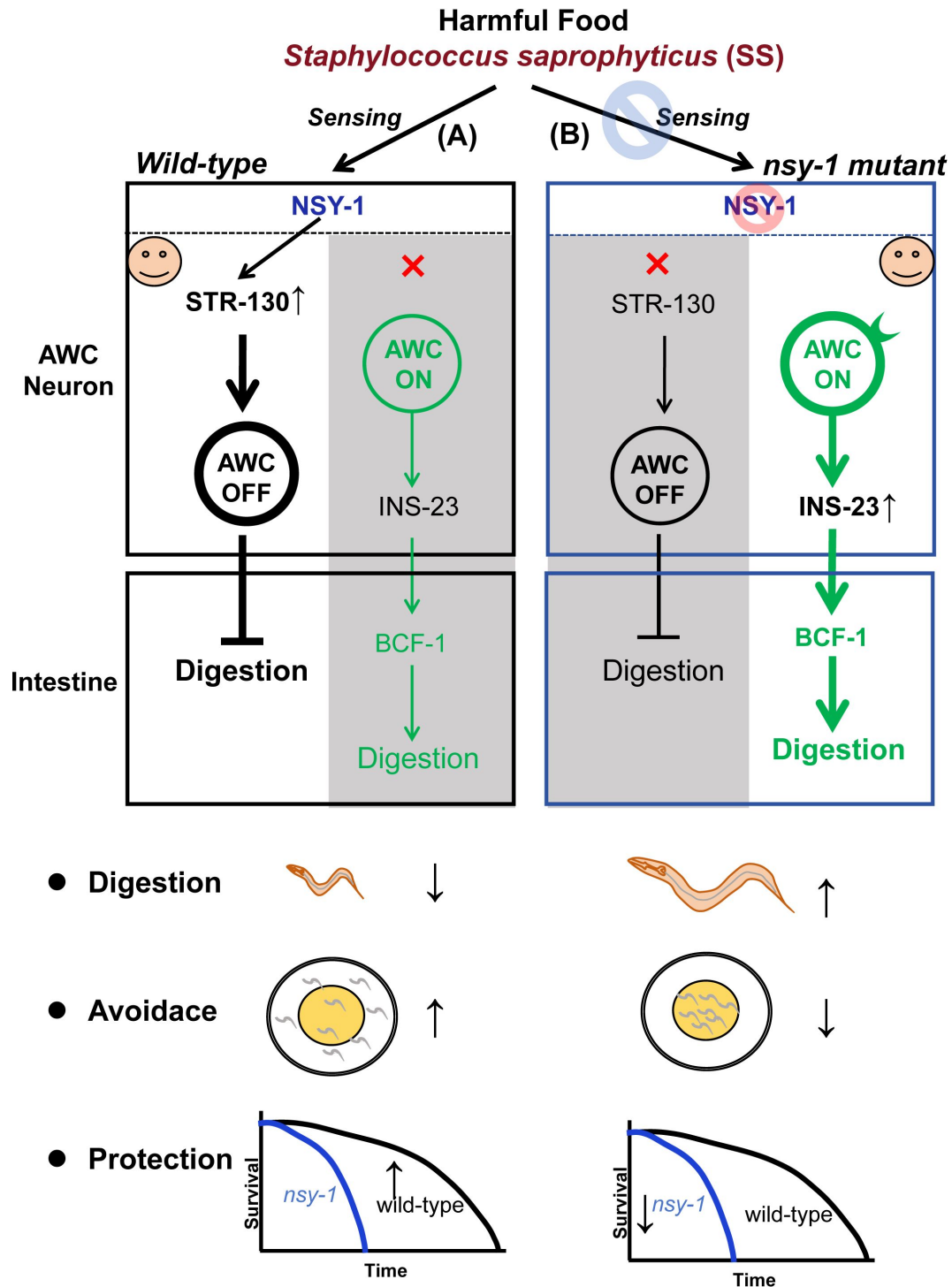


Figure 7.

A model reveals a neural-digestive mechanism for evaluating harmful food.

(A) AWC neuron-expressed NSY-1 detects *Staphylococcus saprophyticus* (SS) as harmful food and shuts down digestion by inducing the AWC^{OFF} neural circuit and NSY-1-dependent STR-130. This mechanism protects animals and helps them avoid harmful food.

(B) Mutations in NSY-1 lead to SS digestion by activating the insulin/IGF-1 signaling (IIS) pathway and BCF-1 expression, thereby reducing the animals' ability to avoid harmful food and decreasing their protection.

humans, providing a foundation for future research on food safety, toxin avoidance, and the neural regulation of digestive processes. This has important implications for public health, as it illuminates the biological mechanisms underlying foodborne illness and the body's defenses against such threats.

Materials and methods

Resource Availability

Lead contact

Further information and requests for reagents may be directed to the Lead contact Bin Qi (qb@yun.edu.cn).

Materials availability

All reagents and strains generated by this study are available through request to the lead contact with a completed Material Transfer Agreement.

Data and code availability

This paper does not report original code.

Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request (qb@ynu.edu.cn).

Experimental model and subject details

C. elegans strains and maintenance

Nematode stocks were maintained on nematode growth medium (NGM) plates seeded with bacteria (*E. coli* OP50) at 20°C.

The following strains/alleles were obtained from the Caenorhabditis Genetics Center (CGC) or as indicated:

1. The following strains were obtained from CGC:

N2 Bristol (wild-type control strain);

AU3: *nsy-1(ag3)*;

ZD101: *tir-1(qd4)*;

RB1971: *bcf-1(ok2599)*;

KU25: *pmk-1(km25)*;

KU4: *sek-1(km4)*;

PY7502: *oyIs85[ceh-36p::TU#813 +ceh-36p::TU#814 + srtx-1p::GFP + un c-122p::DsRed]*

CB1370: *daf-2(e1370)*; shared from mintie Pu lab

CX3695: *str-2::gfp+lin-15(+)*; shared from Huanhu Zhu lab

CX4998: *str-2::gfp+lin-15(+);nsy-1(ky397)*; shared from Hongyun Tang lab

1. The following strains were obtained from published papers.

PHX4067: [*Pbcf-1::bcf-1::gfp::3xflag*](He et al., 2023 [DOI](#));

1. The following strains were constructed by this study:

YNU186: *yIfEx124[Pnsy-1::gfp;Podr-1::rfp]* was constructed by injecting plasmid *Pnsy-1::nsy-1::gfp* with *Podr-1::rfp* in N2 background;

YNU189: *Pbcf-1::bcf-1::gfp::3xflag;nsy-1(ag3)* was constructed by crossing PHX 4067[*Pbcf-1::bcf-1::gfp::3xflag*] with AU3[*nsy-1(ag3)*];

YNU238: *yIf6(nsy-1, EMS mutant)*;

YNU465: *yIfEx252[Podr-1::nsy-1::gfp;Podr-1::rfp;nsy-1(ag3)]* was constructed by injecting plasmid *Podr-1::nsy-1::gfp* with *Podr-1::rfp* in *nsy-1(ag3)* background;

YNU466: *yIfEx253[Pvha-6::nsy-1::gfp;Podr-1::rfp;nsy-1(ag3)]* was constructed by injecting plasmid *Pvha-6::nsy-1::gfp* with *Podr-1::rfp* in *nsy-1(ag3)* background;

YNU488: *yIfEx259[Pstr-130::str-130::mcherry;Podr-1::rfp;nsy-1(ag3)]* was constructed by injecting plasmid *Pstr-130::str-130::mcherry* with *Podr-1::rfp* in *nsy-1(ag3)* background;

YNU491: *yIf57*, AWC neuron specific knock out *nsy-1* strain was constructed by injecting plasmid *pDD162[Podr-1::Cas9+Pu6::nsy-1-sg]*, *nsy-1* repair template (synthesis from Tsingke), *pDD162[Peft-3::Cas9+ Pu6::dpy-10-sg]*, *dpy-10* repair template(synthesis from Tsingke) in PHX4067(*Pbcf-1::bcf-1::gfp::3xflag*) background;

YNU501: *nsy-1(ag3);bcf-1(ok2599)* double mutant was constructed by crossing RB1971[*bcf-1(ok2599)*] with AU3[*nsy-1(ag3)*];

YNU508: *yIfEx266[Pstr130::str-130::mcherry;Podr-1::rfp;nsy-1(ag3)]* was constructed by injecting plasmid *Pstr130::str-130::mcherry* with *Podr-1::rfp* in *str-2::gfp+lin-15(+)* background;

YNU517: *nsy-1(ag3);daf-2(e1370)* double mutant was constructed by crossing CB1370[*daf-2(e1370)*] with AU3[*nsy-1(ag3)*];

YNU729: *yIfEx351[Pges-1::daf-2::gfp;Podr-1::rfp;nsy-1(ag3);daf-2(e1370)]* was constructed by injecting plasmid *Pges-1::daf-2::gfp* with *Podr-1::rfp* in *nsy-1(ag3);daf-2(e1370)* background;

YNU732: *yIfIs48[Podr-1::nsy-1::gfp;Podr-1::rfp;nsy-1(ag3)]* was constructed by injecting plasmid *Podr-1::nsy-1::gfp* with *Podr-1::rfp* in *nsy-1(ag3)* background; YNU730: *yIfEx352[Pceh-36::sid-1::mcherry;pRF4(rol-6);Pbcf-1::bcf-1::gfp::3xflag; sid-1(qt9)]* was constructed by injecting plasmid *Pceh-36::sid-1::mcherry* with *pRF4(rol-6)* in *Pbcf-1::bcf-1::gfp::3xflag;sid-1(qt9)* background.

YNU731: *yIfEx353[Pins-23::ins-23::gfp;Podr-1::rfp]* was constructed by injecting plasmid *Pins-23::ins-23::gfp* with *Podr-1::rfp* in N2 background;

Bacteria Strains

E. coli-OP50 (from Caenorhabditis Genetics Center (CGC)), and *Staphylococcus. Saprophyticus* (from ATCC) were cultured at 37°C in LB medium. A standard overnight cultured bacteria was then spread onto each Nematode growth media (NGM) plate.

Method Details

Generation of transgenic strains

- 1) To construct the *C. elegans* plasmid for expression of *nsy-1*, 1527bp promoter of *nsy-1* was inserted into the pPD95_77-gfp vector. DNA plasmid mixture containing *Pnsy-1::GFP* (20ng/ul) and *Podr-1p::RFP*(50ng/ul) was injected into the gonads of adult wild-type N2 animals.
- 2) To construct the *C. elegans* plasmid for expression of *nsy-1* in AWC neuron, 1348bp promoter of *odr-1* and genomic DNA of *nsy-1* was inserted into the pPD49.26-gfp vector. DNA plasmid mixture containing *Podr-1::nsy-1::GFP* (20ng/ul) and *Podr-1::RFP* (50ng/ul) was injected into the gonads of adult *nsy-1(ag3)*.
- 3) To construct the *C. elegans* plasmid for expression of *str-130*, 2000bp promoter of *str-130* and 1324bp genomic DNA of *str-130* was inserted into the pPD49.26-mcherry vector. DNA plasmid mixture containing *Pstr-130::str-130::mcherry*(20ng/ul) and *Podr-1::rfp* (50ng/ul) was injected into the gonads of adult CX3695[*str-2::gfp+lin-15(+)*].
- 4) To construct the *C. elegans* plasmid for expression of *daf-2* in intestine, 2549bp promoter of *ges-1* and 5400bp cDNA of *daf-2* was inserted into the pPD95_77-gfp vector. DNA plasmid mixture containing *Pges-1::daf-2::gfp*(20ng/ul) and *Podr-1::rfp* (50ng/ul) was injected into the gonads of adult *nsy-1(ag3);daf-2(e1370)*.
- 5) To construct the *C. elegans* plasmid for expression of *sid-1* in AWC neuron, 2000bp promoter of *ceh-36* and 2328bp cDNA of *sid-1* was inserted into the pPD49.26-mcherry vector. DNA plasmid mixture containing *Pceh-36::sid-1::mcherry*(20ng/ul) and *rol-6* (50ng/ul) was injected into the gonads of adult *Pbcf-1::bcf-1::gfp::3xflag;sid-1(qt9)*.
- 6) To construct the *C. elegans* plasmid for expression of *ins-23*, 2017bp promoter of *ins-23* and 286bp genomic DNA of *ins-23* was inserted into the pPD49.26-gfp vector. DNA plasmid mixture containing *Piins-23::ins-23::gfp*(20ng/ul) and *Podr-1::rfp* (50ng/ul) was injected into the gonads of adult wild-type N2 animals.

Generation *nsy-1* AWC neuron specific knock out strain and Genotyping

To construct the *C. elegans* plasmid for knock out of *nsy-1* in AWC neuron, 600bp promoter of *eft-3* was replaced by 1348bp promoter of *odr-1* and *nsy-1* sgRNA was also inserted into the same CRISPR-Cas9-sgRNA vector pDD162 (Dickinson et al., 2013 [DOI](#)). The Cas9 target sites were designed via CRISPR design tool (<http://crispor.tefor.net/> [DOI](#)) and the sgRNA sequences was 5'-GAATTACGCGTTCGAGAAATGG-3'. Knockout strains were generated by injecting 25ng/ul Cas9-sgRNA plasmid, 2μM repair template, co-injection markers include 20ng/ul *dpy-10* Cas9-sgRNA plasmid and 2μM *dpy-10* repair template.

Worms were picked into 10 μl of worm lysis buffer (50 mM KCl, 10 mM Tris-HCl pH 8.0, 2.5 mM MgCl₂, 0.45% NP40, 0.45% Tween-20, 0.01% Gelactin, 0.2 mg/mL Proteinase K), quickly freeze-thaw three times using liquid nitrogen, incubated it at 60°C for 90min and 95 °C for 20min. 1μl supernatant was taken and performed for PCR analysis with the following primers:

nsy-1: forward 5'-CAAGAGGCAAGTGCAGCATA-3', reverse 5'-TGAAGTGTCCATGCTCTCAC-3', then digested with NheI endonuclease overnight and identified by DNA agarose electrophoresis.

Preparation of *Staphylococcus. Saprophyticus* (SS)

SS preparation was followed by our published protocol (Geng et al., 2022 [DOI](#); Liu and Qi, 2023 [DOI](#)). Briefly, a standard overnight culture of SS (37°C in LB broth) was diluted into fresh LB broth (1:100ratio). SS was then spread onto each NGM plate when the diluted bacteria grew to OD600 = 0.5.

Analysis of worm's growth in SS bacteria

The standard overnight cultured SS was then spread onto 60mm NGM plate. Worms were grown on *E. coli*-OP50, and eggs were collected by bleaching and then washing in M9 buffer. Synchronized L1 larvae were obtained by allowing the eggs to hatch in M9 buffer for 12 hr. Synchronized L1s were seeded to plates prepared for the specific assay and incubated at 20°C for 4 days. The developmental conditions were determined by body length.

Food behavior assay

1) Food avoidance assay

Food avoidance assay was performed following to our published methods (Qi et al., 2017 [DOI](#)). Briefly, 5ul of overnight cultured bacteria was seeded on the center of 60mm NGM plates. About 30-50 synchronized L1 animals were seeded onto the bacterial lawns and cultured at 20°C for 8hr. The aversion index was determined by $N(\text{out of lawn})/N(\text{total})$.

2) Food choice assay

Food choice assay was performed following to our published methods (Qi et al., 2017 [DOI](#)). Briefly, 5ul of overnight cultured bacteria was seeded on the different side of 60mm NGM plates. About 300 synchronized L1 animals were seeded onto the center of NGM plates and cultured at 20°C for 8hr. The food trend index was determined by $N(\text{selecting food1})/[N(\text{selecting food1})+N(\text{selecting food2})]$ and $N(\text{selecting food2})/[N(\text{selecting food1})+N(\text{selecting food2})]$.

Lifespan analysis

Larval Lifespan

Studies of larval lifespan were performed on NGM plates at 20°C as previously described (Cui et al., 2013 [DOI](#)). Briefly, L1 staged worms were placed NGM plates or SS seeded NGM plates. Worms were scored every day. Prism8 software was used for statistical analysis.

Adult Lifespan

Studies of lifespan were performed on NGM plates at 20°C as previously described (Kimura et al., 1997 [DOI](#)).

Briefly, lifespan was begun on day zero by placing healthy L4 stage hermaphrodites onto OP50 seeded NGM plates. Animals were transferred to a fresh OP50 or SS seeded plates during the reproductive period (approximately the first ten days) to eliminate self-progeny and every 2 days thereafter. Worms were scored every day. Prism8 software was used for statistical analysis.

EMS mutagenesis

Synchronized L1 animals were grown to the L4 stage on OP50 and then subjected to a 4-hour treatment with 0.5% EMS (Ethyl Methanesulfonate). After treatment, the P0 generation animals were thoroughly washed with M9 solution and transferred to OP50 plates to grow and produce the

F1 generation. Groups of 3-4 F1 animals were picked and placed onto new OP50 plates (3-4 F1 worms per plate) to generate the F2 generation through self-fertilization. Once the F2 generation reached adulthood, all the F2 animals were bleached to obtain the next generation of eggs. Synchronized L1 larvae were obtained by allowing the eggs to hatch in M9 buffer for 12 hours. The synchronized L1s were then seeded onto SS plates and incubated at 20°C for 4 days. Candidate mutants that could grow on SS plates were identified. To confirm the suspected mutants, each candidate mutant was picked singly onto OP50 for passaging and growth to adulthood. These adult animals were then bleached to obtain synchronized L1 mutants. The synchronized L1 mutants were seeded onto SS plates to confirm their ability to grow on SS.

Identification of EMS mutants

DNA isolation, library construction, and whole genome sequencing with gene identified were carried out according to the published protocol (Joseph et al., 2018 [DOI](#)).

i) Preparation of samples for Whole-Genome-Sequencing (WGS)

For each backcross, wild-type males were crossed to mutant hermaphrodites, F1 cross-progeny animals were individually cloned and F2 variants and wild-types were isolated under SS feeding condition. The variant strains and wild-type strains were mixed respectively to constitute the “DNA-pool” used as samples for WGS.

Whole-genome sequencing of pooled F2 recombinants, homozygous for the mutant phenotype following two outcrosses to wild-type N2 animals, was performed to identify the mutations.

ii) WGS data processing

For Whole-Genome-Sequencing, paired-end libraries were sequenced on an Illumina HiSeq 2000. Fastqc was used to control the quality of raw data and Trimmomatic was used to filter the data. Bwa was used to construct the index of *C.elegans* genome and align clean reads to the reference gene sequence (Species: *Caenorhabditis elegans*; Source: UCSC; Reference Genome Version: WBcel235/ce11). The Samtools was used for file format conversion and sorting. Picard was used to remove the duplicate reads and then GATK was used to identify the intervals and realign. The realigned sequence was piled up by Samtools and then inputted to Varscan to call the variants including SNPs and INDELs. Vcfliib was used to perform subtraction between the wild-types and variants. The file was annotated by Snpeff and the candidate genes was finally obtained.

Preparation of samples for RNA sequencing

RNA-seq was done with three biological replicates that were independently generated, collected, and processed. Adult wild type (N2) or *nsy-1* mutant worms were bleached and then the eggs were incubated in M9 for 18 hours to obtain synchronized L1 worms. Synchronized L1 worms were cultured in the NGM plate seeded with OP50 or SS for 4hrs at 20°C. L1 worms were then collected for sequencing.

Fluorescence microscopy of *C. elegans*

Slides for imaging were prepared by making a fresh flattened 5% agarose pad. Worms were mounted on 5% agar pads in M9 buffer with 5mM levamisole then sealed beneath a 22×22mm coverglass. Imaging was done using an Olympus BX53 microscope with a DP80 camera. *str-2::gfp* or *bcf-1::gfp* expression was measured through imaging.

Microscopy

The fluorescence photographs were taken by Olympus BX53 microscope with a DP80 camera. Development statistics were taken by Olympus MVX10 dissecting microscope with a DP80 camera.

Quantification and statistical analysis

Quantification

Animals were randomly selected for fluorescent photography. The size of transgene worms was photographed by the Nomarski microscope and measured by ImageJ software. ImageJ software was used for quantifying fluorescence intensity of indicated animals, which was then normalized with control group.

Statistical analysis

All experiments were performed independently at least three times with similar results.

All statistical analyses were performed using the unpaired two-tailed Student's test. Statistical parameters are presented as mean $\pm$ SD, statistical significance ($P < 0.05$, *; $P < 0.01$, **; $P < 0.001$, ***), and "n" (the number of worms counted).

The Log rank (Mantel-Cox) test was used for lifespan assay and the exact p values of statistics for all survival assays are listed in the Figure.

Acknowledgements

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Additional information

Author Contributions

Y. L and G.T performed experiments, analyzed data and wrote the paper; Z. W performed phenotype analysis; J. Z performed EMS screen; H. L analyzed EMS sequencing data; S. Z supervised some experiments; Z. S supervised some experiments, wrote and revised this paper; B.Q. supervised this study, and wrote the paper with inputs from G. T and Y. L.

Additional files

Supplemental figures. Figure 1—figure supplement 1. SS is harmful food that animals cannot digest. (A) Schematic drawing, microscopic images, and quantitative data from the food choice assay. L1 worms were placed at the center spot (origin). OP50 (yellow) and heat-killed OP50 (blue) bacteria were positioned on opposite sides of the plate. The red point indicates the position of each worm. The percentage of worms on each spot was calculated at the indicated times. Data are represented as mean $\pm$ SD. Scale bar= 1000 μ m. **** p < 0.0001; *** p < 0.001; ** p < 0.01 by Student's t-test. (B) Schematic drawing, microscopic images, and quantitative data from the food choice assay. L1 worms were placed at the center spot (origin). Heat-killed OP50 (blue) and SS (red) bacteria were positioned on opposite sides of the plate. The red point indicates the position of each worm. The percentage of worms on each spot was calculated at the indicated times. Data are represented as mean $\pm$ SD. Scale bar= 1000 μ m. **** p < 0.0001; n.s. not significant by Student's t-test. (C) Schematic drawing and quantitative data of the lifespan of animals fed with SS or OP50. L1 worms were seeded on plates with no food (NGM) or SS bacteria to measure lifespan. n.s. not significant by log-rank test. (D) Schematic drawing and quantitative data of the developmental progression of wild-type N2 under different feeding conditions. L1 animals were seeded onto OP50 plates and grown to the L2 stage. L2 animals were then transferred to OP50, SS, or no food (NGM) plates to measure worm length at the indicated time points. All data are representative of at least three independent experiments. **Figure 2—figure supplement 1. EMS screen to identify genes involved in shutting down digestion of SS.** (A) Schematic illustration of the EMS screen strategy to identify "Y" genes involved in shutting down digestion after sensing SS. In mutants with defects in "Y" genes, digestion of SS is restored, allowing the mutants to grow on SS. (B) Developmental phenotype of wild-type N2 and ylf6 mutant worms fed with SS bacteria. Scale bar = 200 μ m. (C) Schematic drawing showing the mutation sites in nsy-1(ylf6) and nsy-1(ag3). **Figure 3—figure supplement 1. Construction of nsy-1 specific knockout in AWC neurons using CRISPR-Cas9.** Representative DNA gels showing NheI digestion of PCR-amplified genomic DNA extracted from wild-type (WT) worms and worms with nsy-1-specific knockout in AWC neurons (odr-1p::Cas9 + u6p::nsy-1-sg). **Figure 3—figure supplement 2. NSY-1 functions in intestine to shut down SS Digestion.** (A) Microscopic image showing the expression pattern of nsy-1. nsy-1 is mainly expressed in the head neurons and intestine. Scale bar = 50 μ m. (B) Developmental progression of nsy-1(ag3) mutant worms carrying Pvha-6::nsy-1::gfp (intestine-specific expression) grown on SS bacteria. Control animals are labeled with white stars, and animals carrying the transgenes (rescued animals) are labeled with yellow stars. Data are represented as mean $\pm$ SD. Scale bar = 200 μ m. **** p < 0.0001 by Student's t-test. n= number of animals which were scored. **Figure 3—figure supplement 3. NSY-1 functions in AWC neurons to influences the recognition of SS.** (A) Microscopic images, and quantitative data of the food dwelling/avoidance assay. The animals were scored at the indicated times after L1 worms were placed on the food spot. The blue circle indicates the edge of the bacterial lawn, and the red point indicates the position of each worm. Data are represented as mean $\pm$ SD. Scale bar= 500 μ m. * p < 0.05 by Student's t-test. (B) Microscopic images, and quantitative data of the food dwelling/avoidance assay. The animals were scored at the indicated times after L1 worms were placed on the food spot. The blue circle indicates the edge of the bacterial lawn, and the red point indicates the position of each worm. Data are represented as mean $\pm$ SD. Scale bar = 500 μ m. ** p < 0.01; * p < 0.05 by Student's t-test. (C) Microscopic images, and quantitative data from the food choice assay. L1 worms were placed at the center spot (origin). Heat-killed OP50 and SS bacteria were positioned on opposite sides of the plate. The red point indicates the position of each worm. The percentage of worms on each spot was calculated at the indicated times. Data are represented as mean $\pm$ SD. Scale bar= 2 mm. **** p < 0.0001; *** p < 0.001; ** p < 0.01 by Student's t-test. (D) Microscopic images, and quantitative data from the food choice assay. L1 worms were placed at the center spot (origin). Heat-killed OP50 and SS bacteria were positioned on opposite sides of the plate. The red point indicates the position of each worm. The percentage of worms on each spot was calculated at the indicated times. Data are represented as mean $\pm$ SD. Scale bar = 2 mm. ** p < 0.01; * p < 0.05; n.s. not significant by Student's t-test. **Figure 3—figure supplement 4. AWC neurons are essential for initiating SS digestion.** Developmental phenotype of wild-type N2 and AWC(-) worms fed with SS bacteria. Data are represented as mean $\pm$ SD. Scale bar =

500 μm . n.s. not significant by Student's t-test. n= number of animals which were scored. **Figure 4—figure supplement 1. str-130 is induced in wild-type in response to SS, dependent on NSY-1.**(A) GO enrichment analysis of 304 NSY-1-dependent candidate genes responding to SS (Supplementary file 2). Genes related to sensory perception (sra-32, str-87, str-112, str-130, str-160, str-230) are highlighted as enriched (red arrow). (B) Relative mRNA expression levels of sensory perception-related genes (sra-32, str-87, str-112, str-130, str-160, str-230) extracted from RNA-seq data. These genes are induced in wild-type N2 animals in response to SS, but their expression is reduced in nsy-1(ag3) mutant animals under SS feeding conditions, indicating that the induction of these genes in wild-type in response to SS is dependent on NSY-1. **Figure 4—figure supplement 2. str-130 as the dominant effector of NSY-1-mediated SS response regulation.**(A) Developmental progression of wild-type animals treated with control RNAi, str-130 RNAi, sra-32 RNAi, str-230 RNAi, str-87 RNAi or str-160 RNAi grown on SS bacteria. Data are represented as mean $\pm$ SD. Scale bar = 200 μm . ****p < 0.0001; ***p < 0.001; *p < 0.05; n.s. not significant by Student's t-test (candidate RNAi vs Control RNAi). n= number of animals which were scored. (B) Developmental progression of nsy-1(ag3) mutant animals treated with control RNAi, str-130 RNAi, sra-32 RNAi, str-230 RNAi, str-87 RNAi, str-112 RNAi or str-160 RNAi grown on SS bacteria. Data are represented as mean $\pm$ SD. Scale bar=500 μm . n.s. not significant by Student's t-test (candidate RNAi vs Control RNAi). n= number of animals which were scored. **Figure 4—figure supplement 3. NSY-1 inhibits SS digestion by inducing GPCR str-130.** Developmental progression of wild-type or nsy-1(ag3) mutant animals treated with control RNAi or str-130 RNAi grown on SS bacteria. Data are represented as mean $\pm$ SD. Scale bar = 200 μm . ****p < 0.0001; n.s. not significant by Student's t-test. n= number of animals which were scored. **Figure 5—figure supplement 1. nsy-1 mutation induces the expression of insulin-related genes.**(A) GO enrichment analysis of 308 genes induced by the nsy-1 mutation under SS feeding conditions (also see Supplementary file 2). (B) Relative mRNA expression levels of insulin-related genes (ins-23, ins-22, ins-27, ins-24) extracted from RNA-seq data. These genes are upregulated in nsy-1(ag3) mutant animals under SS feeding conditions, which may contribute to SS digestion in the nsy-1 mutant. **Figure 5—figure supplement 2. NSY-1 mutation promotes SS digestion by inducing ins-23.** Developmental progression of nsy-1(ag3) mutant animals treated with control RNAi, ins-22 RNAi, ins-23 RNAi, ins-24 RNAi, or ins-27 RNAi grown on SS bacteria. Data are represented as mean $\pm$ SD. Scale bar = 500 μm . ****p < 0.0001; n.s. not significant by Student's t-test (candidate RNAi vs Control RNAi). n= number of animals which were scored. **Figure 5—figure supplement 3. INS-23 induction in nsy-1 mutants promotes digestion independently of intestinal DAF-2 function.**(A) CeNGEN predicts showing the head neurons expression pattern of ins-23. ins-23 is expressed in AWC neurons. (B) Developmental progression of nsy-1(ag3);daf-2(e1370) mutant worms carrying Pges-1::daf-2::gfp (intestine-specific expression) grown on SS bacteria. Control animals are labeled with white stars, and animals carrying the transgenes (rescued animals) are labeled with yellow stars. Data are represented as mean $\pm$ SD. Scale bar = 200 μm . ****p < 0.0001 by Student's t-test. n= number of animals which were scored. (C) Microscopic images and quantitative data showing fluorescence of Pins-23::ins-23::GFP animals treated with control RNAi or pmk-1 RNAi. Data are represented as mean $\pm$ SD. Scale bar = 10 μm . n.s. not significant by Student's t-test. n= number of animals which were scored. **Figure 6—figure supplement 1. INS-23 and STR-130 are functions in AWC neurons to affect BCF-1 expression.**(A) Microscopic images and quantitative data showing fluorescence of Pbcf-1::bcf-1::GFP animals treated with control RNAi or AWC ins-23 RNAi. Data are represented as mean $\pm$ SD. Scale bar= 100 μm . ****p < 0.0001 by Student's t-test. n= number of animals which were scored. (B) Microscopic images and quantitative data showing fluorescence of Pbcf-1::bcf-1::GFP animals treated with control RNAi or AWC str-130 RNAi. Data are represented as mean $\pm$ SD. Scale bar= 100 μm . **p < 0.01 by Student's t-test. n= number of animals which were scored. [🔗](#)

Supplemental file 1. List of genes induced by SS food that are dependent on NSY-1. Related to Figure 4B. [🔗](#)

Supplemental file 2. GO enrichment analysis of 304 NSY-1-dependent candidate genes responding to SS. Related to Figure 4—figure supplement 1A and B. [🔗](#)

Supplemental file 3. List of genes induced in *nsy-1(ag3)* mutant animals feeding on SS. Related to Figure 5B. [↗](#)

Supplemental file 4. GO enrichment analysis of 308 genes induced by the *nsy-1* mutation under SS feeding conditions. Related to Figure 5—figure supplement 1A and B. [↗](#)

Note

This reviewed preprint has been updated to correct a corresponding author's email address.

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Reviewer #1 (Public review):

Summary:

In this manuscript, Liu et al have tried to dissect the neural and molecular mechanisms that *C. elegans* use to avoid the digestion of harmful bacterial food. Liu et al show that *C. elegans* use ON-OFF state of AWC olfactory neurons to regulate the digestion of harmful gram-positive bacteria *S. saprophyticus* (SS). Authors show that when *C. elegans* are fed on SS food, AWC neurons switch to OFF fate, which prevents the digestion of *S. saprophyticus*, and this helps *C. elegans* avoid these harmful bacteria. Using genetic and transcriptional analysis as well as making use of previously published findings, Liu et al implicate p38 MAPK pathway (in particular, NSY-1, the *C. elegans* homolog of MAPKKK ASK1) and insulin signaling in this process.

Strengths:

The revised manuscript has improved significantly. The authors have addressed almost all the comments that I had in my initial review.

Weaknesses:

None.

<https://doi.org/10.7554/eLife.104028.2.sa3>

Reviewer #2 (Public review):

Summary:

Using *C. elegans* as a model, the authors present an interesting story demonstrating a new regulatory connection between olfactory neurons and the digestive system. Mechanistically, they identified key factors (NSY-1, STR-130 et.al) in neurons, as well as critical 'signaling factors' (INS-23, DAF-2) that bridge different cells/tissues to execute the digestive shutdown induced by poor-quality food (*Staphylococcus saprophyticus*, SS).

Strengths:

The conclusions of this manuscript are mostly well supported by the experimental results shown.

Weaknesses:

The authors have done a nice job in addressing my comments.

<https://doi.org/10.7554/eLife.104028.2.sa2>

Reviewer #3 (Public review):

Summary:

The study explores a molecular mechanism by which *C. elegans* detects low-quality food through neuron-digestive crosstalk, offering new insights into food quality control systems.

Liu and colleagues demonstrated that NSY-1, expressed in AWC neurons, is a key regulator for sensing *Staphylococcus saprophyticus* (SS), inducing avoidance behavior and shutting down the digestive system via intestinal BCF-1. They further revealed that INS-23, an insulin peptide, interacts with the DAF-2 receptor in the gut to modulate SS digestion. The study uncovers a food quality control system connecting neural and intestinal responses, enabling *C. elegans* to adapt to environmental challenges.

Strengths:

The study employs a genetic screening approach to identify *nsy-1* as a critical regulator in detecting food quality and initiating adaptive responses in *C. elegans*. The use of RNA-seq analysis is particularly noteworthy, as it reveals distinct regulatory pathways involved in food sensing (Figure 4) and digestion of *Staphylococcus saprophyticus* (Figure 5). The strategic application of both positive and negative data mining enhances the depth of analysis. Importantly, the discovery that *C. elegans* halts digestion in response to harmful food and employs avoidance behavior highlights a physiological adaptation mechanism.

Weaknesses:

Major weaknesses have been addressed.

<https://doi.org/10.7554/eLife.104028.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

In this manuscript, Liu et al have tried to dissect the neural and molecular mechanisms that C. elegans use to avoid digestion of harmful bacterial food. Liu et al show that C. elegans use the ON-OFF state of AWC olfactory neurons to regulate the digestion of harmful gram-positive bacteria S. saprophyticus (SS). The authors show that when C. elegans are fed on SS food, AWC neurons switch to OFF fate which prevents digestion of S. saprophyticus and this helps C. elegans avoid these harmful bacteria. Using genetic and transcriptional analysis as well as making use of previously published findings, Liu et al implicate the p38 MAPK pathway (in particular, NSY-1, the C. elegans homolog of MAPKKK ASK1) and insulin signaling in this process.

Strengths:

The authors have used multiple approaches to test the hypothesis that they present in this manuscript.

Weaknesses:

Overall, I am not convinced that the authors have provided sufficient evidence to support the various components of their hypothesis. While they present data that loosely align with their hypothesis, they fail to consider alternative explanations and do not use rigorous approaches to strengthen their overall hypothesis. The selective picking of genes from the RNA sequencing data and forcing the data to fit the proposed hypothesis based on previously published findings, without exploring other approaches, indicates a lack of thoroughness and rigor. These critical shortcomings significantly diminish enthusiasm for the manuscript in its totality. In my opinion, this is the biggest weakness in this manuscript.

We appreciate the reviewer's all the suggestions which help us to improve this paper. We now addressed reviewer's comments at the section of "Reviewer #1 (Recommendations for the authors)"

Reviewer #2 (Public review):

Summary:

Using C. elegans as a model, the authors present an interesting story demonstrating a new regulatory connection between olfactory neurons and the digestive system.

Mechanistically, they identified key factors (NSY-1, STR-130 et.al) in neurons, as well as critical 'signaling factors' (INS-23, DAF-2) that bridge different cells/tissues to execute the digestive shutdown induced by poor-quality food (Staphylococcus saprophyticus, SS).

Strengths:

The conclusions of this manuscript are mostly well supported by the experimental results shown.

Weaknesses:

Several issues could be addressed and clarified to strengthen their conclusions.

(1) The word "olfactory" should be carefully used and checked in this manuscript. Although AWCs are classic olfactory neurons in C. elegans, no data in this manuscript supports the idea that olfactory signals from SS drive the responses in the digestive system. To validate that it is truly olfaction, the authors may want to check the responses of worms (e.g. AWC, digestive shutdown, INS-23 expression) to odors from SS.

We appreciate the reviewer's careful attention to terminology. We agree that the term "olfactory" requires direct experimental validation. However, in this paper, we only used "olfactory" to specific define the AWC neurons. As reviewer's suggestion, we now deleted the word "olfactory".

(2) In line 113, what does "once the digestive system is activated" mean? The authors need to provide a clearer statement about 'digestive activation' and 'digestive shutdown'.

Previously, we observed that activating larval digestion with heat-killed E. coli or E. coli cell wall peptidoglycan (PGN) enabled the digestion of SS as food (Hao et al., 2024). Additionally, when animals reached the L2 stage by feeding normal OP50 diet, they could utilize SS as a food source to support growth (Figure 1—figure supplement 1D). These findings suggest that once digestion is activated (via E. coli components or L2-stage maturation), worms gain the capacity to process SS as a viable food source, abolishing SS-induced growth impairment (Hao et al., 2024) (Figure 1—figure supplement 1D).

(3) No control data on OP50. This would affect the conclusions generated from Figures 2A, 2B, 2D, 3B, 3C, 3G, 4D-G, 5D-E, 6B-D.

We appreciate this point. The central goal of the experiments listed (Figures 2A,B,D; 3B,C,G; 4D-G; 5D-E; 6B-D) was not to compare growth or behavior between SS and OP50 under standard conditions, but rather to understand the genetic basis of the C. elegans response specifically to SS, as identified through our nsy-1 mutant screen.

Our data in Figure 1 clearly establishes the fundamental difference in growth and feeding behavior when larvae encounter SS compared to OP50 (Figures 1A,B). Having established SS

as an unfavorable food source that triggers a specific protective response (digestive shutdown), the subsequent experiments focus on deciphering how this response is mediated.

Therefore, within these specific experimental contexts under SS feeding: The primary comparison is between wild-type (N2) and *nsy-1* mutant animals. All assays (growth, behavior, survival) are performed under the same SS feeding conditions for both genotypes.

This design allows us to directly assess the functional role of NSY-1 in mediating the SS-specific response pathway we are investigating. Including an OP50 control for every figure would not address this core genetic question and could introduce confounding variables given the established difference in how *C. elegans* treats these two food sources. The critical internal control for these specific experiments is the performance of the wild-type under SS versus the mutant under SS.

(4) Do the authors know which factors are released from AWC neurons to drive the digestive shutdown?

Enrichment analysis revealed that genes related to extracellular functions, such as insulin-related genes, are induced in *nsy-1* mutant animals (Figure 5—figure supplement 1A, Supplementary file 4). Further analysis of insulin-related genes from the RNA-seq data showed that *ins-23* is predominantly induced in *nsy-1* mutant animals (Figure 5—figure supplement 1B), suggesting its potential role in promoting SS digestion. We found that knockdown of *ins-23* in *nsy-1* mutants inhibited SS digestion (Figure 5D). Given that *INS-23* is expressed in AWC neurons (Figure 5—figure supplement 3A, CeNGEN), this suggests increased production and likely enhanced release of *INS-23* from AWC neurons in the *nsy-1* mutant background, which promotes SS digestion.

The insulin/insulin-like growth factor signaling (IIS) pathway, particularly through the DAF-2 receptor, integrates nutritional signals to regulate various behavioral and physiological responses related to food (Kodama et al., 2006; Ryu et al., 2018). It has been shown that *INS-23* acts as an antagonist for the DAF-2 receptor to promote larval diapause (Matsunaga et al., 2018). To test whether *ins-23* induction in *nsy-1* mutants promotes SS digestion through its receptor, DAF-2, we constructed a *nsy-1; daf-2* double mutant. We found that the SS digestion ability of the *nsy-1* mutant was inhibited by the *daf-2* mutation. This suggests that the *nsy-1* mutation induces the insulin peptide *ins-23*, which promotes SS digestion through its potential receptor, DAF-2.

The data supports a model where AWC neurons regulate digestion via the release of *INS-23*. Loss of *nsy-1* function increases *INS-23* release from AWC, activating DAF-2 signaling and promoting digestion. Conversely, in wild-type animals, reduced *INS-23* release from AWC contributes to digestive shutdown in response to SS food.

Reviewer #3 (Public review):

Summary:

The study explores a molecular mechanism by which C. elegans detects low-quality food through neuron-digestive crosstalk, offering new insights into food quality control systems. Liu and colleagues demonstrated that NSY-1, expressed in AWC neurons, is a key regulator for sensing Staphylococcus saprophyticus (SS), inducing avoidance behavior and shutting down the digestive system via intestinal BCF-1. They further revealed that INS-23, an insulin peptide, interacts with the DAF-2 receptor in the gut to modulate SS digestion. The study uncovers a food quality control system connecting neural and intestinal responses, enabling C. elegans to adapt to environmental challenges.

Strengths:

The study employs a genetic screening approach to identify nsy-1 as a critical regulator in detecting food quality and initiating adaptive responses in C. elegans. The use of RNA-seq analysis is particularly noteworthy, as it reveals distinct regulatory pathways involved in food sensing (Figure 4) and digestion of Staphylococcus saprophyticus (Figure 5). The strategic application of both positive and negative data mining enhances the depth of analysis. Importantly, the discovery that C. elegans halts digestion in response to harmful food and employs avoidance behavior highlights a physiological adaptation mechanism.

Weaknesses:

Major points:

(1) While NSY-1 positively regulates str-130 expression in AWC neurons and is critical for SS avoidance and survival, the authors should examine whether similar phenotypes are observed in str-130 mutants.

In this study, we mainly focused on how worms sense adverse food sources (SS food) and shutdown digestion (not growth as digestion shutdown readout). We found that nsy-1 in AWC play key roles in response SS food, once nsy-1 mutation, mutant animals cannot detect SS food and digest it, therefore growth under SS food. From RNA-seq, we found that nsy-1 positively regulates several sensory perception related genes (sra-32, str-87, str-112, str-130, str-160, str-230) (Figure 4—figure supplement 1A, Supplementary file 2). After screen, we found that we found that knockdown of str-130 in wild-type animals promoted SS digestion, thereby supporting animal growth (Figure 4D), and the proportion of animals with two AWC^{OFF} neurons decreased (Figure 4E). Secondly, we found that overexpression of str-130 in nsy-1 mutant animals inhibited SS digestion, thereby slowing animal growth (Figure 4F), and the proportion of animals with two AWC^{OFF} neurons increased (Figure 4G). These results demonstrate that NSY-1 promotes the AWC^{OFF} state by inducing str-130 expression, which in turn inhibits SS digestion in C. elegans.

(2) NSY-1 promotes the AWC-OFF state through str-130, inhibiting SS digestion. The authors should investigate whether STR-130 in AWC neurons regulates bcf-1 expression levels in the intestine.

We agree with the reviewer's suggestion regarding the potential role of STR-130 in AWC neurons regulating intestinal bcf-1 expression. To address this, we generated transgenic worms with AWC-specific knockdown of str-130, achieved by rescuing sid-1 cDNA expression under the ceh-36 promoter (AWC-specific) in sid-1(qt9);BCF-1::GFP background worms.

We observed that AWC neuron-specific RNAi of str-130 elevated intestinal BCF-1::GFP expression (Figure 6—figure supplement 1B). This demonstrates that STR-130 functions cell-non-autonomously in AWC neurons to repress BCF-1 expression in the intestine.

(3) The current results rely on str-2 expression levels to indicate the AWC state. Ablating AWC neurons and testing the effects on digestion would provide stronger evidence for their role in digestive regulation.

To confirm the important of AWC state in SS digestion, we performed AWC-specific neuron ablation experiments using previously validated transgenic strain that expresses cleaved caspase under the AWC-specific promoter, ceh-36 (ceh-36p::caspase). Critically, worms with ablated AWC neurons completely failed to digest SS food (Figure 3—figure supplement 4), phenocopying the non-digesting state of wild-type worms on SS when AWC-OFF signaling is

impaired. This result directly confirms that functional AWC neurons are essential for initiating SS digestion, aligning with our model where the AWC-OFF state (induced by SS) inhibits digestion while the AWC-ON state promotes it.

Furthermore, we previously study discovered that AWC ablation activates the intestinal mitochondrial unfolded protein response and inhibits food digestion, mechanistically linking neuronal integrity to gut stress responses and digestive inhibition.

Together, these functional ablation studies provide compelling physiological evidence that AWC neurons act as central regulators of food-state sensing and gut function.

(4) The claim that NSY-1 inhibits INS-23 and that INS-23 interacts with DAF-2 to regulate bcf-1 expression (Line 339-340) requires further validation. Neuron-specific disruption of INS-23 and gut-specific rescue of DAF-2 should be tested.

We agree with the reviewer that the proposed NSY-1 → INS-23 → DAF-2 → BCF-1 signaling axis requires tissue-specific validation. To address this, we conducted compartment-specific functional dissection of INS-23 and DAF-2:

AWC neuronal role of INS-23:

To test whether INS-23 acts in AWC neurons to regulate intestinal BCF-1, we generated AWC-specific knockdown strains which was achieved by rescuing *sid-1* cDNA expression under the *ceh-36* promoter in a *sid-1(qt9);BCF-1::GFP* background. We found that AWC-restricted *ins-23* knockdown significantly reduced intestinal BCF-1::GFP expression (Figure 6—figure supplement 1A). This confirms that INS-23 functions cell-non-autonomously within AWC sensory neurons to activate intestinal BCF-1, consistent with NSY-1's upstream inhibition of INS-23 in this neuronal subtype

Intestinal role of DAF-2 as INS-23 receptor:

To investigate whether DAF-2 acts as the gut-localized receptor for neuronal INS-23 signaling, we performed tissue-specific rescue experiments in the *nsy-1(ag3);daf-2(e1370)* double mutant. When DAF-2 was re-introduced specifically in the intestine (using the *ges-1* promoter), we observed a significant suppression of SS digestion (Figure 5—figure supplement 3B), but not rescue digestive defect. This indicates that INS-23 induction in *nsy-1* mutants promotes digestion independently of intestinal DAF-2 function.

(5) Figure Reference Errors: Lines 296-297 mention Figure 6E, which does not exist in the main text. This appears to refer to Figure 5E, which has not been described.

We corrected this.

Reviewer #1 (Recommendations for the authors):

I would like the authors to address the following comments in a resubmission.

(1) The hallmark of the activated p38 MAPK pathway is the phosphorylation of most downstream kinase p38 (PMK-1/PMK2 in C. elegans) of this kinase cascade. Previous work from Bergmann lab showed that the most downstream kinase of this pathway, PMK-1/PMK-2, is not required for AWC asymmetry. I wonder whether that is the case also for the model that Liu et al have presented in this manuscript. Since p38/PMK-1 undergoes activation (phosphorylation) in response to pathogenic bacteria like P. aeruginosa, it is worth testing whether PMK-1 plays a role downstream of NSY-1 in the model that Liu et al present in this manuscript. It would be worth testing whether there is increased phosphorylation of p38 when C. elegans are fed SS and whether that

phosphorylation regulates downstream components that Liu et al have identified in this manuscript.

We thank the reviewer for raising this important point regarding PMK-1/p38 MAPK signaling. As established in our prior work (Reference 1), SS exposure triggers phosphorylation of PMK-1 (P-PMK-1) in *C. elegans*, and *pmk-1* mutants exhibit enhanced growth on SS (Figure-1, Figure-2). This confirms that PMK-1-mediated innate immune signaling actively regulates SS responsiveness and digestion.

To address whether PMK-1 functions downstream of NSY-1 within our proposed model, we performed critical epistasis analyses. While we observed that *nsy-1* mutation elevates *ins-23* (indicating NSY-1 suppression of *ins-23*), knockdown of *pmk-1* did not alter *ins-23* expression levels (Figure 5-figure supplement 3C). This demonstrates that PMK-1 does not operate through the *ins-23* pathway to regulate SS digestion. Thus, although both pathways respond to SS, the PMK-1-mediated innate immune response and the NSY-1/INS-23 axis constitute distinct regulatory mechanisms governing digestive adaptation.

Reference 1: Geng, S., Li, Q., Zhou, X., Zheng, J., Liu, H., Zeng, J., Yang, R., Fu, H., Hao, F., Feng, Q., & Qi, B. (2022). Gut commensal *E. coli* outer membrane proteins activate the host food digestive system through neural-immune communication. *Cell host & microbe*, 30(10), 1401–1416.e8. <https://doi.org/10.1016/j.chom.2022.08.004>

(2) Since p38 MAPK pathway has a well-established role in host defense in the C. elegans intestine, it is important to show that NSY-1 does not function in the intestine in the model that Liu et al present. I would like the authors to reintroduce nsy-1 in C. elegans intestine in nsy-1 mutant animals and then test whether it has any effect on worm length on SS food (similar to what is done in Figure 3 for AWC-specific nsy-1).

Beyond its established role in AWC neurons, we detected NSY-1 expression in the intestine (Figure 3-figure supplement 2A). To assess intestinal NSY-1 function, we performed tissue-specific rescue experiments in *nsy-1* mutants using the intestinal-specific *vha-1* promoter. Intestinal expression of NSY-1 significantly suppressed the enhanced SS digestion phenotype in *nsy-1* mutants (Figure 3-figure supplement 2B), demonstrating functional involvement of gut-localized NSY-1 in regulating digestive responses. We propose intestinal NSY-1 mediates this effect through innate immune signaling, consistent with its known pathway components. As previously established (Reference 1), the canonical PMK-1/p38 MAPK pathway functions downstream of NSY-1, with both *sek-1* and *pmk-1* knockdown enhancing SS digestion through immune modulation. This indicates intestinal NSY-1 suppresses digestion may act through PMK-1-mediated immune responses. Since neuronal NSY-1's role in digestive control was previously undefined, we prioritized mechanistic analysis of its neuronal function in digestion regulation.

Notably, this immune-mediated mechanism operates independently of NSY-1's neuronal regulation pathway. In AWC neurons, NSY-1 controls digestion exclusively through the neuropeptide signaling axis (INS-23/DAF-2/BCF-1) without engaging innate immune components.

Reference 1: Geng, S., Li, Q., Zhou, X., Zheng, J., Liu, H., Zeng, J., Yang, R., Fu, H., Hao, F., Feng, Q., & Qi, B. (2022). Gut commensal *E. coli* outer membrane proteins activate the host food digestive system through neural-immune communication. *Cell host & microbe*, 30(10), 1401–1416.e8. <https://doi.org/10.1016/j.chom.2022.08.004>

(3) At multiple places, wild-type (WT) controls have been labeled as N2. It is better to label all controls as WT (and not as N2).

Corrected.

(4) In Figure 2B, the aversion response should be scored at multiple time points, like Figure 1C, rather than at just one timepoint.

We thank the reviewer for suggesting multi-timepoint analysis of aversion behavior. In accordance with this recommendation, we have now quantified SS avoidance at multi-timepoint. As shown in the revised Figure 2B, *nsy-1* mutants exhibited significantly impaired avoidance responses at both 4h and 6h but not at 8h, confirming that NSY-1 is essential for sustained aversion to SS food in the early response. This data demonstrates that the critical role of NSY-1 in food discrimination at initial sensory responses.

(5) Does the re-introduction of nsy-1 in AWC neurons in nsy-1 mutant background help animals avoid SS in dwelling and food-choice assays? Along the same lines, does the CRISPR-generated AWC-specific mutant of NSY-1 fail to avoid SS in dwelling and food-choice assays similar to the whole-animal mutant? These behavioral data are missing in Figure 3.

We thank the reviewer for prompting behavioral validation of AWC-specific *nsy-1* functions. To determine whether NSY-1 in AWC neurons mediates SS sensory perception, we performed dwelling (avoidance) and food-choice assays using AWC-specific *nsy-1* knockout and AWC-rescued strains (*nsy-1(ag3); Podr-1::nsy-1*). In dwelling assays, AWC-specific *nsy-1* KO mutants exhibited significantly impaired SS avoidance at 6h (Figure 3-figure supplement 3A), while AWC-rescued strains restored avoidance capacity at 2-6h (Figure 3-figure supplement 3B). Food-choice assays further revealed that AWC *nsy-1* KO mutants preferentially migrated toward SS (Figure 3-figure supplement 3C), whereas AWC-rescued showed no preference between SS and HK-E. coli (Figure 3-figure supplement 3D). These data conclusively demonstrate that NSY-1 acts in AWC neurons to mediate SS recognition and aversion behaviors.

(6) In Figure 3E and F, the number of animals that were used for scoring AWC str-2p::GFP expression should be specified.

we added the number of animals in the figure.

(7) RNA seq analysis identified multiple GPCRs (including STR-130) that are upregulated in an NSY-1-dependent manner when animals are fed with SS bacteria. However, the authors decided to only characterize STR-130 because of previously published findings. It is important to rule out the role of other GPCRs since all are upregulated on SS food as shown in Figure S4 B. I would like the authors to knock down other GPCRs in the same manner as they did for STR-130 and demonstrate that only str-130 knockdown behaves similarly to the nsy-1 mutant (if that is the case) using the assay presented in Figure 4 D.

We appreciate the reviewer's suggestion to comprehensively evaluate NSY-1-regulated GPCRs. In response, we extended our functional analysis to all six GPCRs (*str-130*, *str-230*, *str-87*, *str-112*, *str-160*, and *sra-32*) identified as NSY-1-dependent and SS-induced in RNA-seq (Figure 4—figure supplement 1).

Using RNAi knockdown and the SS growth assay, we observed that RNAi of *str-130*, *str-230*, *str-87*, or *str-112* significantly enhanced SS growth (Figure 4—figure supplement 2A), with *str-130* RNAi exhibiting the most robust phenotype—phenocopying *nsy-1* mutants. Crucially, none of these GPCR knockdowns further enhanced growth in *nsy-1(ag3)* mutants (Figure 4—figure supplement 2B), confirming their position downstream of NSY-1. These data establish

str-130 as the dominant effector of NSY-1-mediated SS response regulation, while suggesting minor contributions from other GPCRs (str-230, str-87, str-112).

(8) In Figure 4E and G, the number of animals that were used for scoring GFP expression should be specified.

we added the number of animals in the figure.

(9) When comparing Figure 3E and Figure 4E, it appears that the loss of str-130 RNAi does not phenocopy nsy-1 mutant. This raises the question of whether the inefficiency of RNAi targeting str-130 is the cause, or if STR-130 is not the only GPCR regulated by NSY-1 on SS food. I would like the authors to address this discrepancy. If RNAi inefficiency is indeed the cause, using an RNAi-sensitive background, such as an eri-1 mutant, could help strengthen the data presented in Figure 4E. Conversely, if RNAi inefficiency is not responsible for the discrepancy, I suggest that the authors investigate the roles of other GPCRs that were identified by RNA sequencing.

We appreciate the reviewer's observation regarding the phenotypic difference between nsy-1 mutants and str-130 (RNAi) animals on SS food (Fig. 3E vs Fig. 4E).

While both genetic perturbations significantly enhance SS growth and increase the proportion of animals exhibiting AWC^{ON} states compared to wild type (indicating enhanced digestion), the specific AWC^{ON} neuron configurations differ: nsy-1 mutants predominantly show 2 AWC^{ON} animals, whereas str-130(RNAi) animals primarily exhibit the 1 AWC^{ON} /1 AWC^{OFF} configuration (Fig. 3E vs Fig. 4E).

This difference likely arises because STR-130 is the key GPCR mediating NSY-1's inhibitory effect on SS digestion, but it is not the sole GPCR involved, as evidenced by our RNAi screen identifying several additional NSY-1-regulated GPCRs (str-230, str-87, str-112) whose depletion also enhanced SS growth (Fig. 4A-D).

The robust SS growth enhancement and AWC^{ON} state increase caused by str-130 (RNAi) (phenocopying the nsy-1 mutant's functional outcome of enhanced digestion) (Figure 4D, 4E) indicate effective RNAi knockdown for this specific assay. Therefore, the distinct neural configurations reflect the partial redundancy among GPCRs downstream of NSY-1, rather than an inherent inefficiency of the str-130 RNAi.

The nsy-1 mutant phenotype represents the complete loss of all inhibitory GPCR signaling coordinated by NSY-1, while str-130(RNAi) represents the loss of its major component. Investigating the roles of other identified GPCRs (str-230, str-87, str-112) in modulating AWC^{ON} neuron states is an important direction for future research.

(10) In Figure 4 F and 4 G, the authors show that the overexpression of STR-130 rescues the nsy-1 mutant phenotype suggesting that NSY-1 might function through STR-130 to control digestion on SS food. These data place STR-130 downstream of NSY-1. To further strengthen these epistasis data, authors should knock down str-130 in nsy-1 mutant animals and show that the combined loss of both genes produces the same effect as the loss of either gene alone.

We thank the reviewer for the insightful suggestion to further define the genetic relationship between nsy-1 and str-130. To strengthen our epistasis analysis, we performed RNAi knockdown of str-130 in the nsy-1(ag3) mutant background and assessed development on SS food. Consistent with STR-130 acting downstream of NSY-1, the loss of str-130 via RNAi did not further enhance the developmental capacity (i.e., growth phenotype) of nsy-1(ag3) mutant animals on SS. This lack of enhancement indicates that str-130 and nsy-1 function within the

same genetic pathway, with str-130 acting epistatically downstream of nsy-1 (Figure 4—figure supplement 3). This finding reinforces the model proposed from our overexpression data (Fig. 4F-G) – that NSY-1 primarily exerts its inhibitory effect on SS digestion by inducing the expression GPCR STR-130.

(11) In Figure 5C, please mention "ins-23 transcript levels" on the top of the graph so that it is clear what these data represent.

We appreciate the reviewer's suggestion.

(12) Since all ins genes were upregulated in nsy-1 mutants (though ins-23 was indeed the most highly upregulated gene) on SS food from RNA seq analysis (Figure S5 B), it is important to first phenotypically characterize all of them using "worm length assay". If this analysis shows that ins-23 has the most robust phenotype, it would make more sense to just focus on ins-23.

We agree with the reviewer that initial phenotypic characterization of candidate genes identified through transcriptomic analysis is valuable. Our RNA-seq data revealed that several insulin-like peptide genes, including ins-22, ins-23, ins-24, and ins-27, were significantly upregulated in the nsy-1 mutant on SS food (Figure 5—figure supplement 1B). We prioritized these insulin-like peptide genes for functional validation because they are known to act as neuropeptides capable of mediating non-cell autonomous signaling in previous studies (Shao et al 2016).

To determine if any were functionally responsible for the enhanced SS growth observed in nsy-1 mutants, we performed functional phenotypic screening using the SS growth assay (worm length assay). We individually knocked down each of these candidates (ins-22, ins-23, ins-24, ins-27) in the nsy-1(ag3) mutant background. Among these, only RNAi targeting ins-23 significantly attenuated (i.e., suppressed) the enhanced development of the nsy-1(ag3) mutant on SS (Figure 5—figure supplement 2). This targeted functional screening revealed that ins-23 has the most robust and specific role in mediating the enhanced digestion phenotype downstream of NSY-1 loss, providing the critical justification for our subsequent focus on this particular insulin-like peptide.

Ref:

Shao, L. W., Niu, R., & Liu, Y. (2016). Neuropeptide signals cell non-autonomous mitochondrial unfolded protein response. Cell research, 26(11), 1182–1196. <https://doi.org/10.1038/cr.2016.118>

Reviewer #2 (Recommendations for the authors):

There are several minor errors and typos in the manuscript

(1) A number of typos in the figures, like "length".

Corrected.

(2) The 'axis labels' are inconsistent from panel to panel, like "relative body length" and "relative worm length".

Corrected.

(3) The fonts are inconsistent from panel to panel.

Corrected.

| (4) *There is no Ex unique number for transgenic lines.*

Corrected.

| **Reviewer #3 (Recommendations for the authors):**

Minor points:

| (1) *Figure 3B, 3C, 3G, 4D, 4F, 5D, 5E, and 6C: Replace "lenth" with "length" (consistent with Figure 2A).*

Corrected.

| (2) *Figure 4D: Correct "ctontrol" to "control."*

Corrected.

| (3) *Figure 4G: Update the co-injection marker to Podr-1::GFP instead of Pstr-2::GFP.*

Corrected.

| (4) *Figure 5C: This figure is missing from the Results section.*

Corrected.

| (5) *Figure 6A: Label the graph with Pbcf-1::bcf-1::GFP, as in Figure 6D.*

Corrected.

| (6) *Italicization: Lines 588 and 603-italicize nsy-1.*

Corrected.

| (7) *Supplementary Figure S2A: Correct "Screeng" to "Screening."*

Corrected.

| (8) *Spelling/Proofreading: Ensure consistent spelling and grammar, such as correcting "mutan" to "mutant" in Figure 4A.*

Corrected.

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